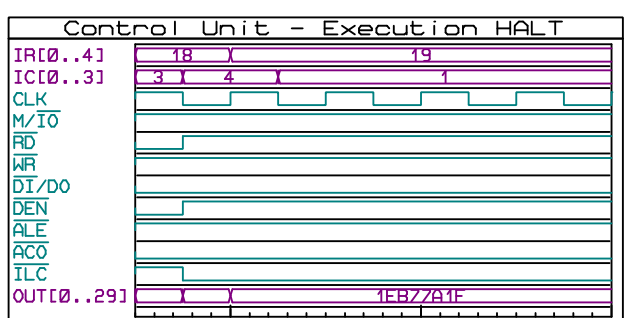
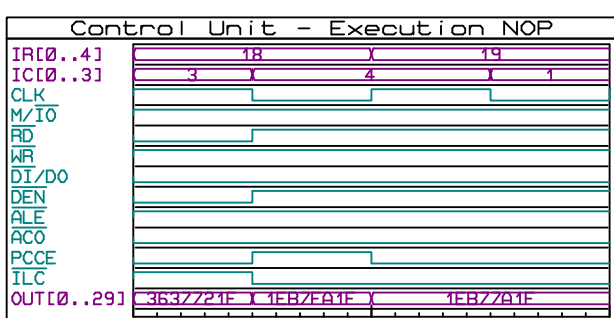
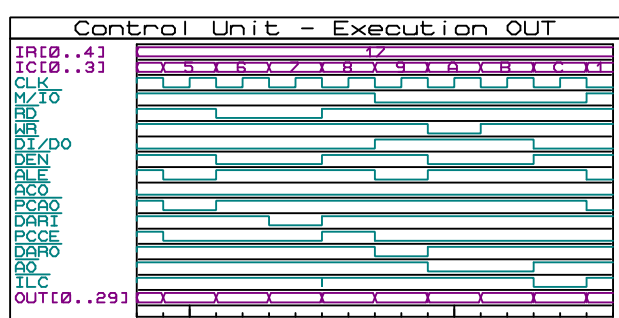
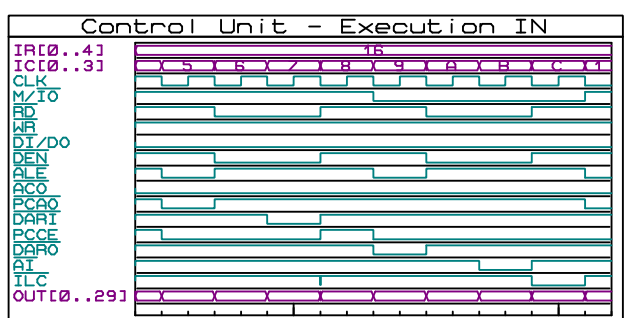
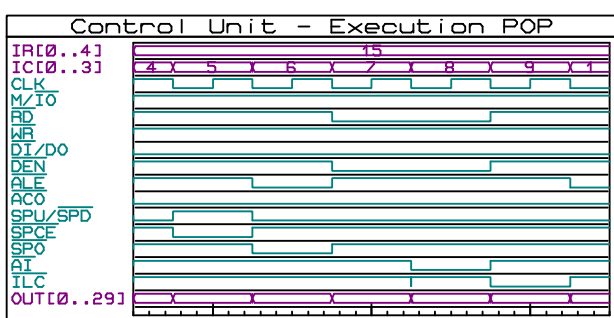
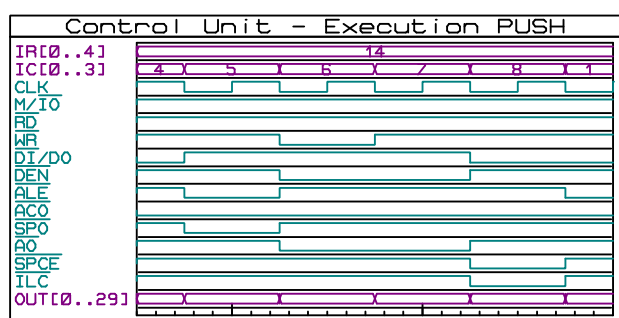
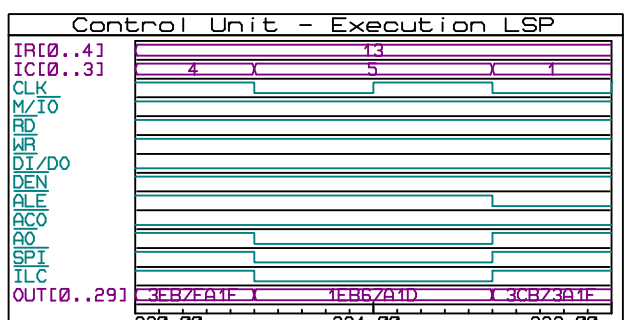
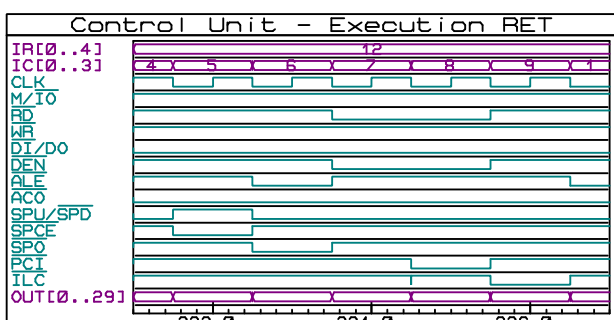
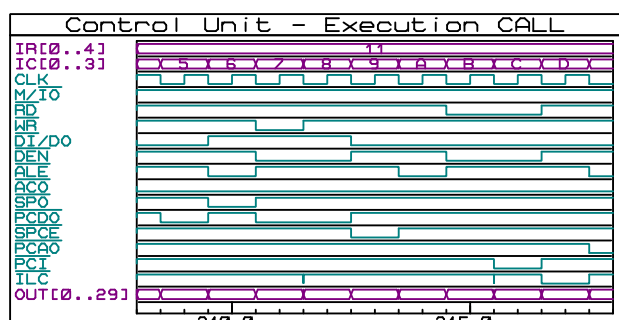
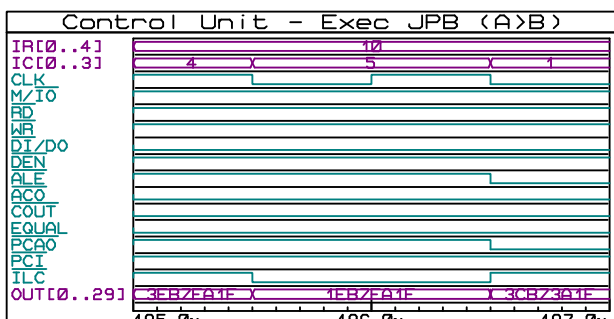
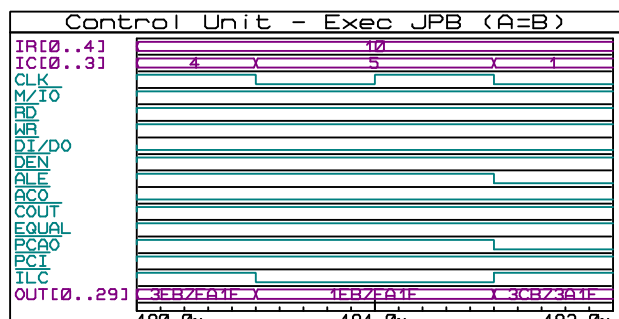
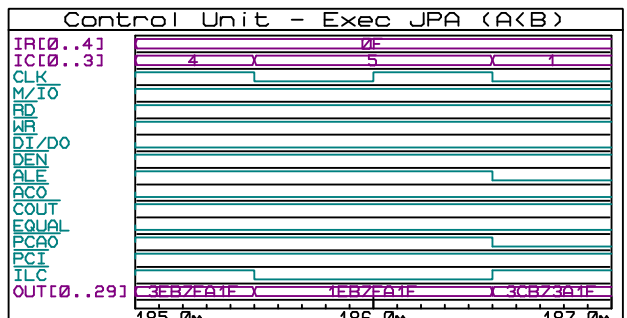
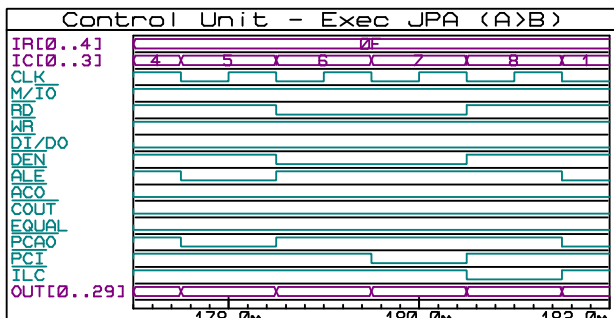
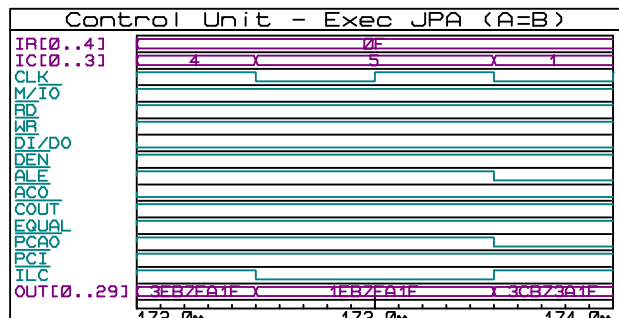
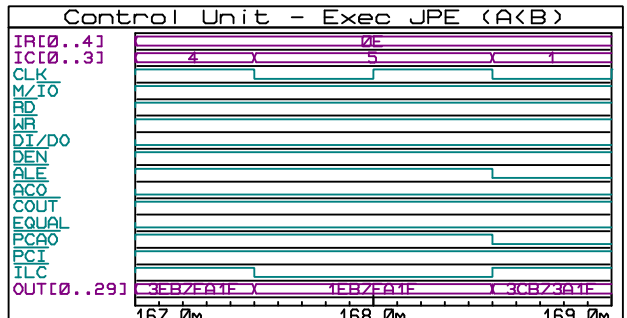
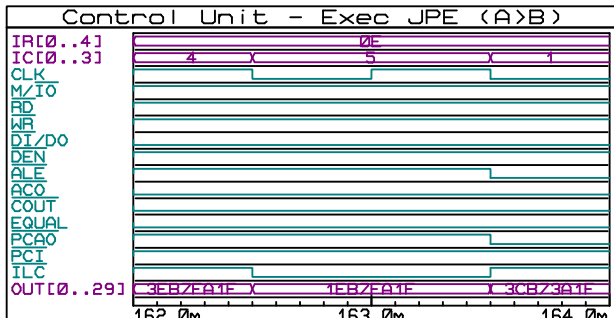
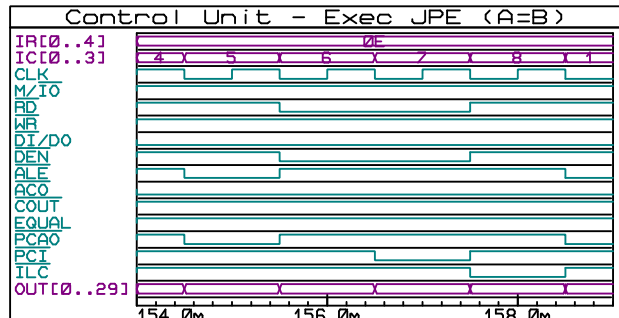
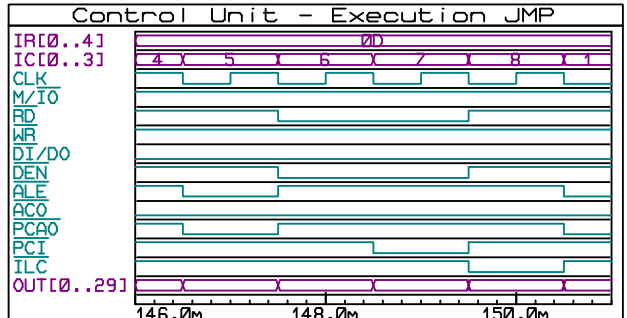
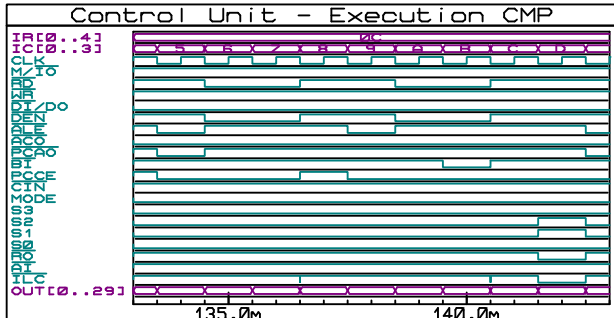
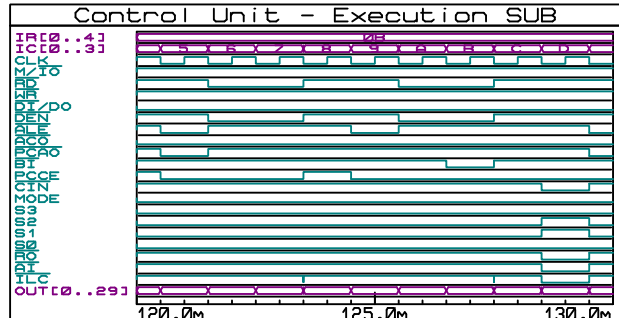
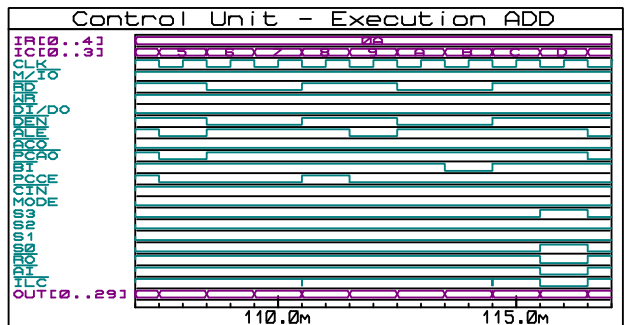
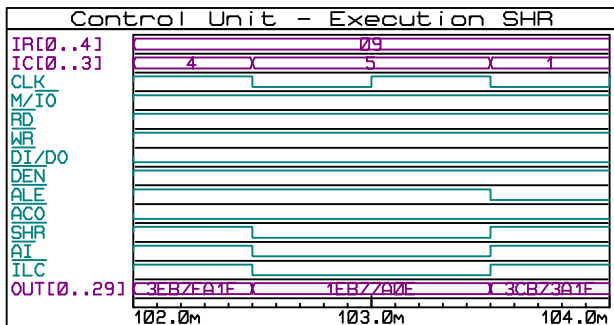
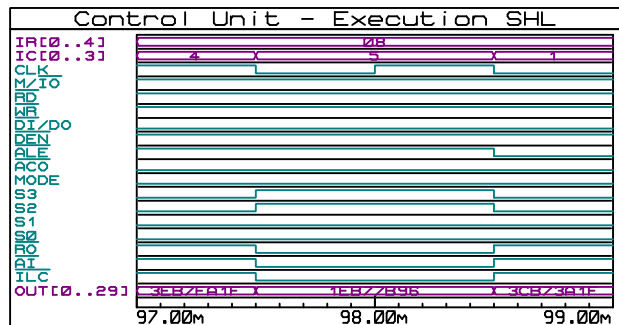
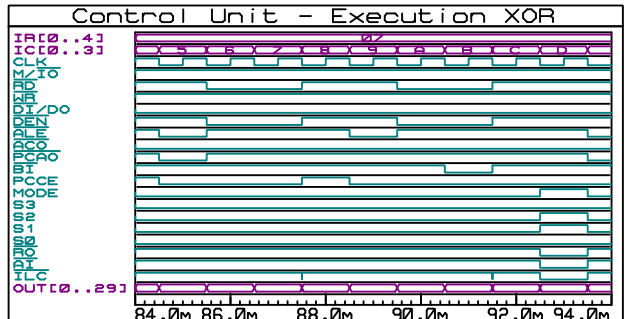
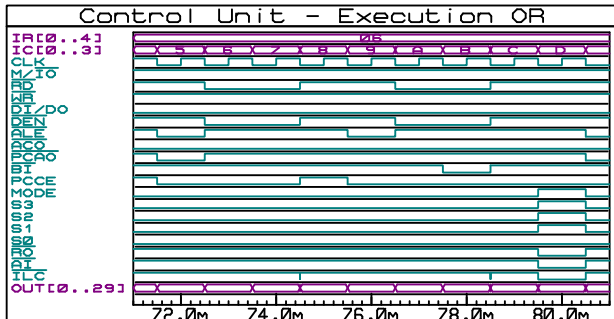
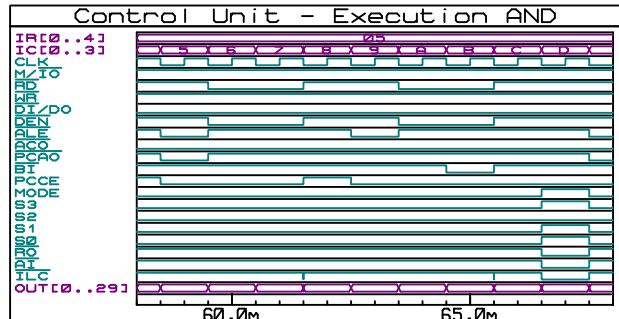
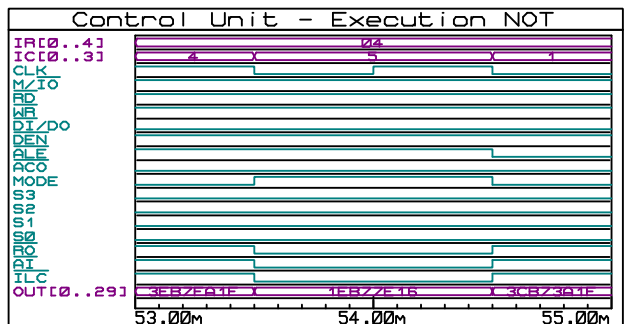
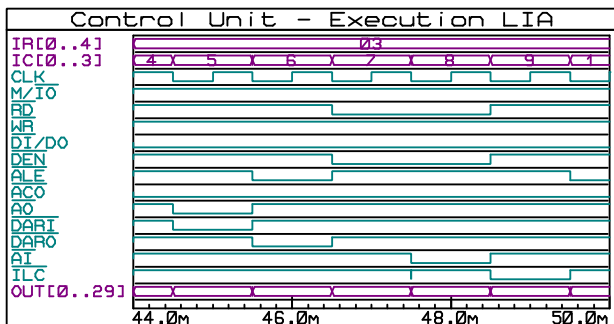
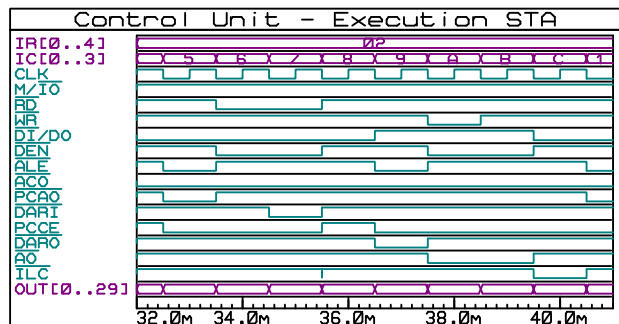
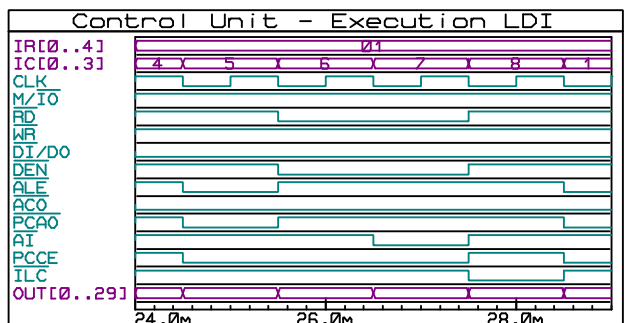
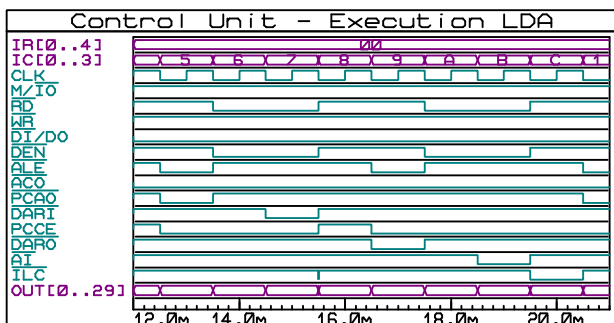
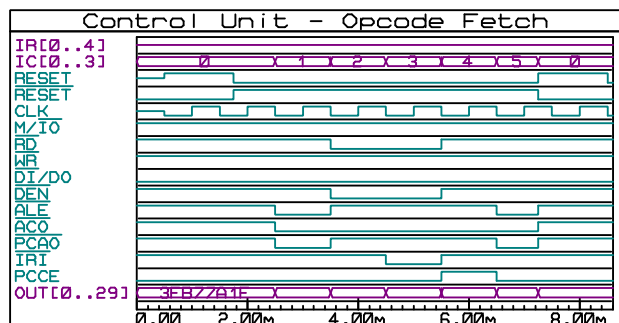
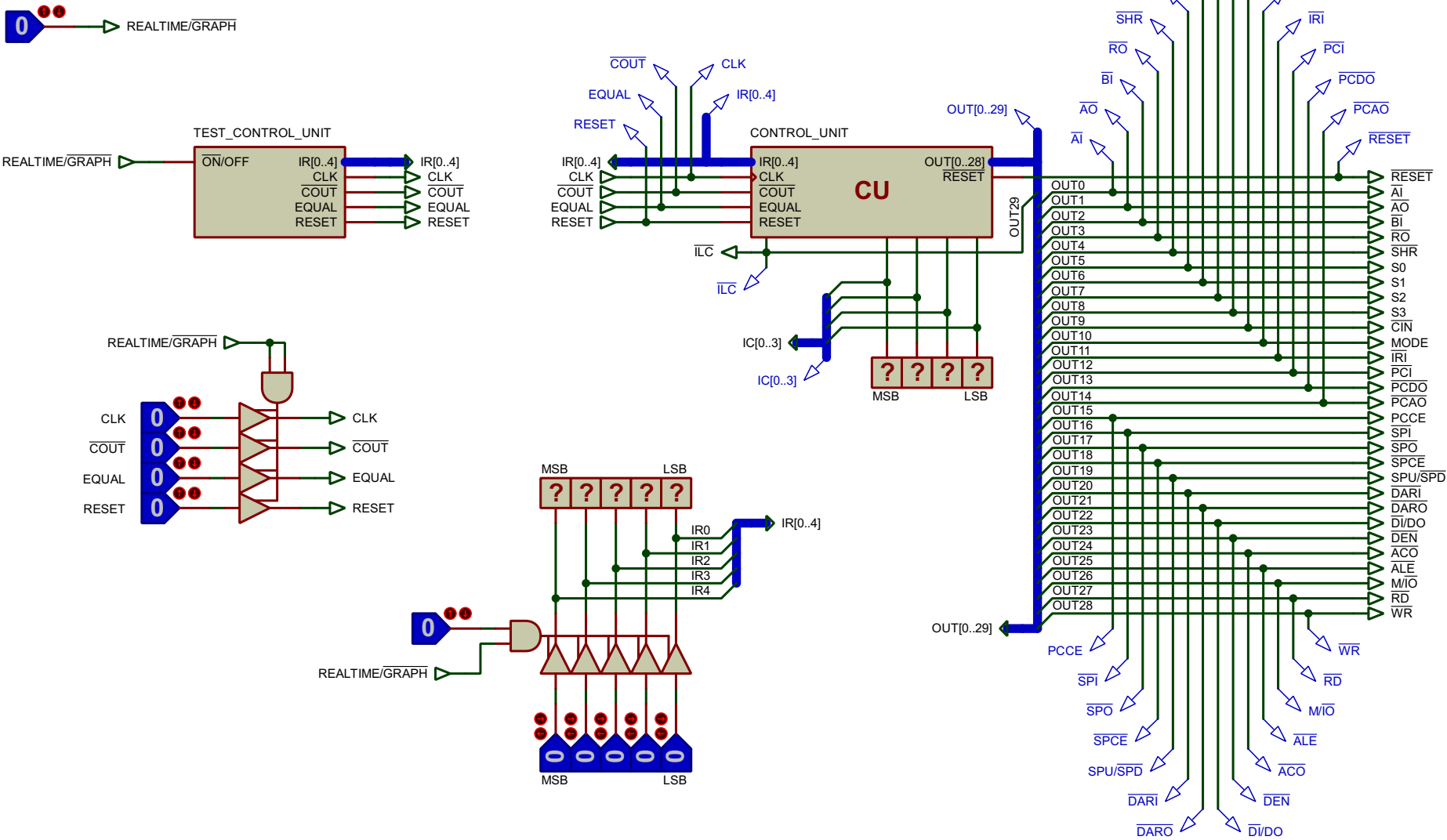
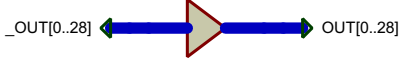
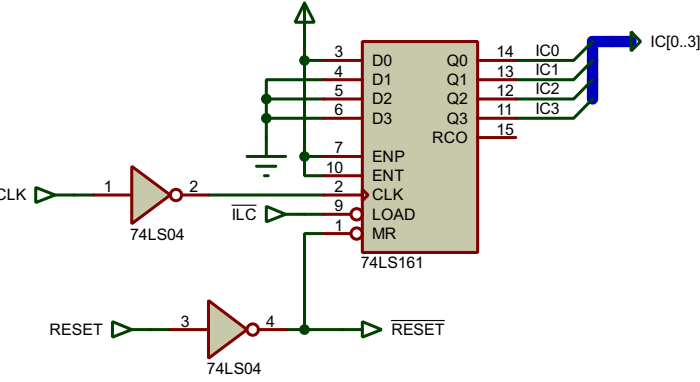
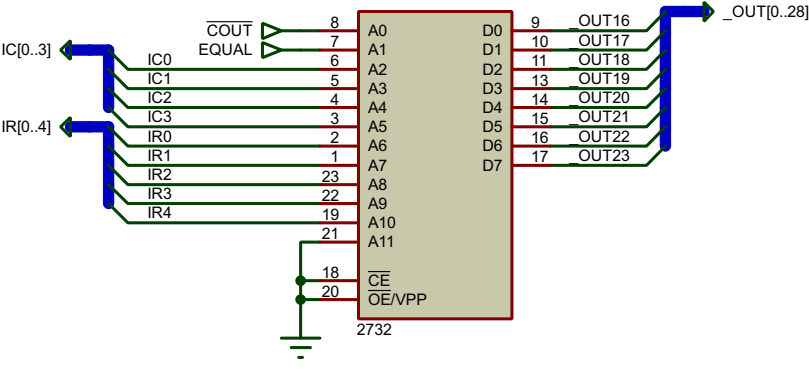
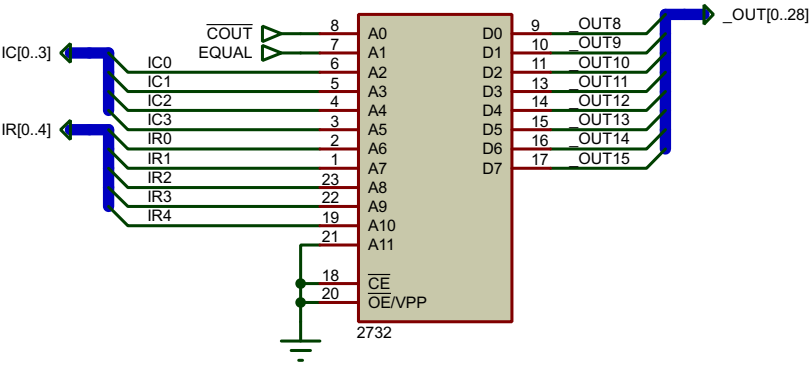
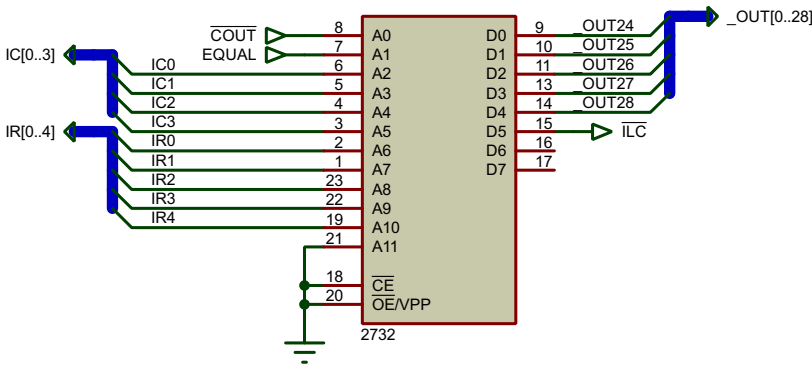
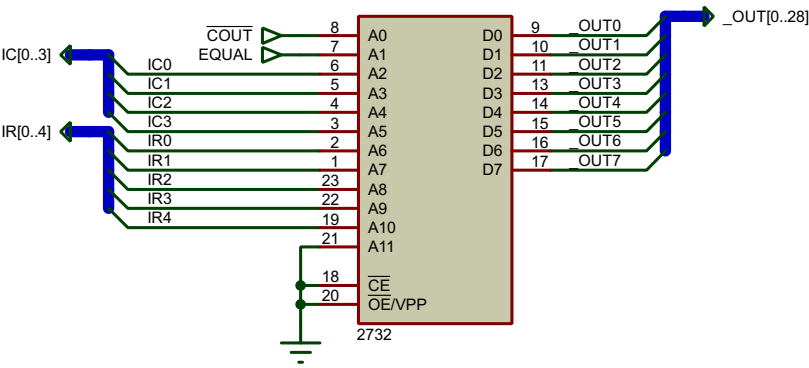




CONTROL_UNIT



TEST_CONTROL_UNIT

*SCRIPT GENERATOR TEST_CONTROL_UNIT

ALIAS ONOFF = \$ON\$/OFF
ALIAS COUT = \$COUT\$

PIN ONOFF
PIN IR[0..4] = FLT
PIN COUT = FLT
PIN EQUAL = FLT
PIN RESET = FLT

INT OP_LDA = \$00, OP_LDI = \$01, OP_STA = \$02, OP_LIA = \$03
INT OP_NOT = \$04, OP_AND = \$05, OP_OR = \$06, OP_XOR = \$07
INT OP_SHL = \$08, OP_SHR = \$09, OP_ADD = \$0A, OP_SUB = \$0B
INT OP_CMP = \$0C, OP_JMP = \$0D, OP_JPE = \$0E, OP_JPA = \$0F
INT OP_JPB = \$10, OP_CALL = \$11, OP_RET = \$12, OP_LSP = \$13
INT OP_PUSH = \$14, OP_POP = \$15, OP_IN = \$16, OP_OUT = \$17
INT OP_NOP = \$18, OP_HALT = \$19

IF ONOFF = TRUE THEN

IR[0..4] = FLT
COUT = FLT
EQUAL = FLT
RESET = FLT

ELSE

RESET = 1 AFTER 0.5m
RESET = 0 AFTER 1.75m
RESET = 1 AFTER 7.25m
RESET = 0 AFTER 8.5m
COUT = 1 AFTER 0.5m
EQUAL = 0 AFTER 0.5m
SLEEP FOR 12m
IR[0..4] = OP_LDA : SLEEP FOR 12m
IR[0..4] = OP_LDI : SLEEP FOR 8m
IR[0..4] = OP_STA : SLEEP FOR 12m
IR[0..4] = OP_LIA : SLEEP FOR 9m
IR[0..4] = OP_NOT : SLEEP FOR 5m
IR[0..4] = OP_AND : SLEEP FOR 13m
IR[0..4] = OP_OR : SLEEP FOR 13m
IR[0..4] = OP_XOR : SLEEP FOR 13m
IR[0..4] = OP_SHL : SLEEP FOR 5m
IR[0..4] = OP_SHR : SLEEP FOR 5m
IR[0..4] = OP_ADD : SLEEP FOR 13m
IR[0..4] = OP_SUB : SLEEP FOR 13m
IR[0..4] = OP_CMP : SLEEP FOR 13m
IR[0..4] = OP_JMP : SLEEP FOR 8m
IR[0..4] = OP_JPE : COUT = 1 : EQUAL = 1 : SLEEP FOR 8m
IR[0..4] = OP_JPE : COUT = 0 : EQUAL = 0 : SLEEP FOR 5m
IR[0..4] = OP_JPE : COUT = 1 : EQUAL = 0 : SLEEP FOR 5m
IR[0..4] = OP_JPA : COUT = 1 : EQUAL = 1 : SLEEP FOR 5m
IR[0..4] = OP_JPA : COUT = 0 : EQUAL = 0 : SLEEP FOR 8m
IR[0..4] = OP_JPA : COUT = 1 : EQUAL = 0 : SLEEP FOR 5m
IR[0..4] = OP_JPB : COUT = 1 : EQUAL = 1 : SLEEP FOR 5m
IR[0..4] = OP_JPB : COUT = 0 : EQUAL = 0 : SLEEP FOR 5m
IR[0..4] = OP_JPB : COUT = 1 : EQUAL = 0 : SLEEP FOR 8m
IR[0..4] = OP_CALL : SLEEP FOR 13m
IR[0..4] = OP_RET : SLEEP FOR 9m
IR[0..4] = OP_LSP : SLEEP FOR 5m
IR[0..4] = OP_PUSH : SLEEP FOR 8m
IR[0..4] = OP_POP : SLEEP FOR 9m
IR[0..4] = OP_IN : SLEEP FOR 12m
IR[0..4] = OP_OUT : SLEEP FOR 12m
IR[0..4] = OP_NOP : SLEEP FOR 4m
IR[0..4] = OP_HALT

ENDIF

END

*ENDSCRIPT

*SCRIPT GENERATOR TEST_CU_CLOCK

ALIAS ONOFF = \$ON\$/OFF

PIN ONOFF
PIN CLK = FLT

INT i

IF ONOFF = TRUE THEN

CLK = FLT

ELSE

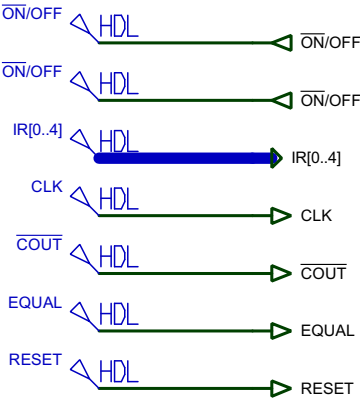
FOR i = 0 TO 300 STEP 1
SLEEP FOR 0.5m
CLK = 0
SLEEP FOR 0.5m
CLK = 1

NEXT i

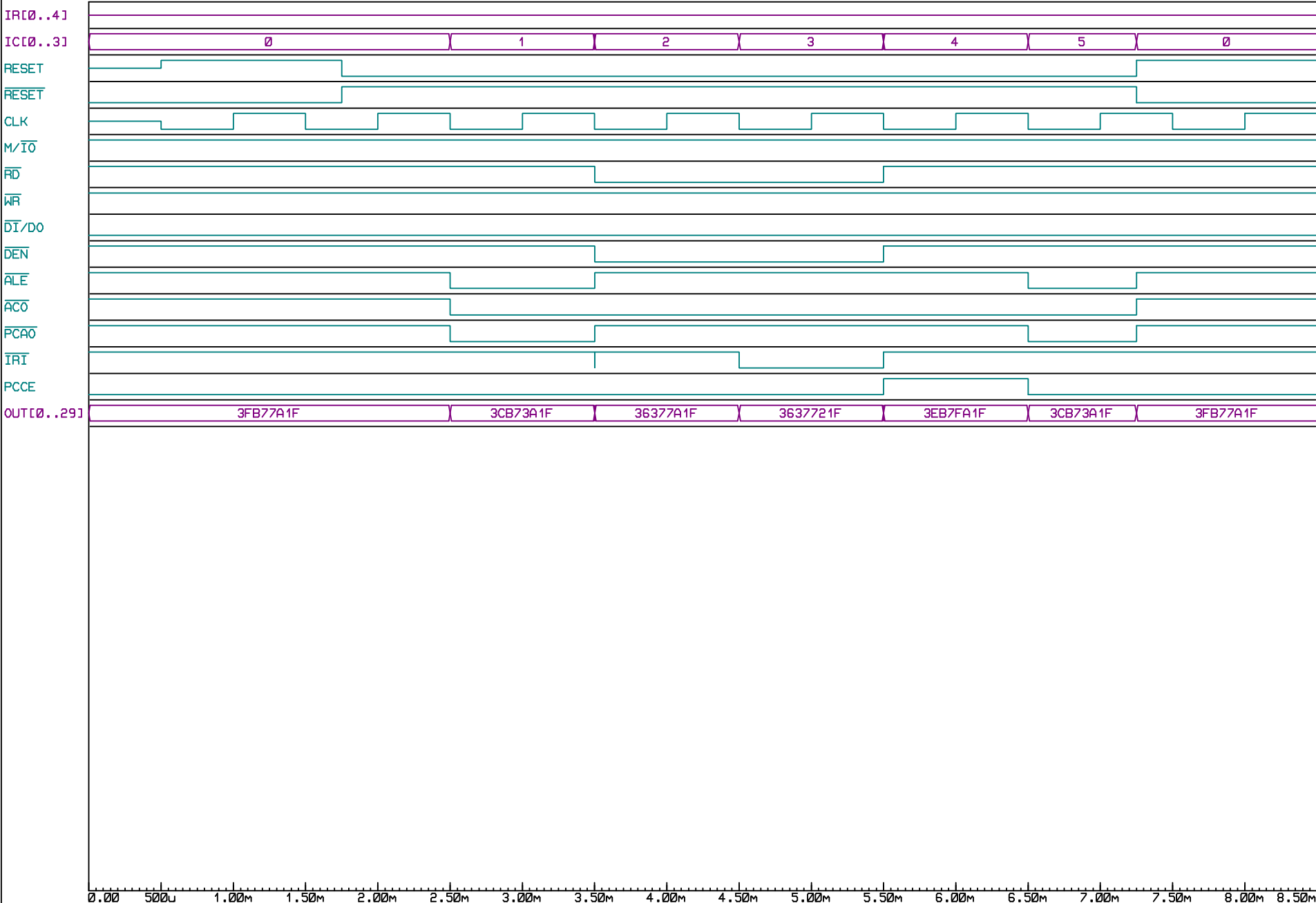
ENDIF

END

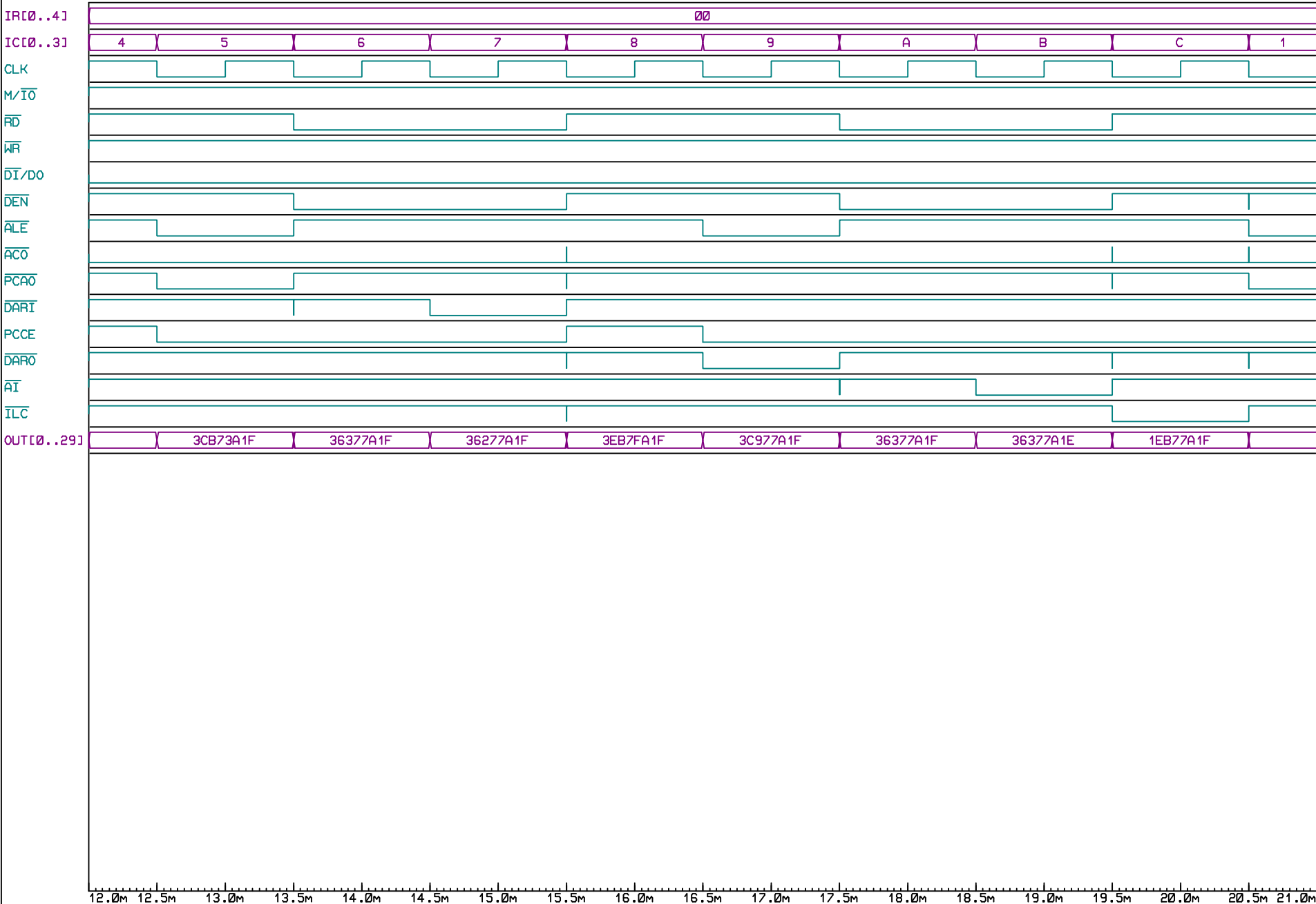
*ENDSCRIPT



Control Unit - Opcode Fetch



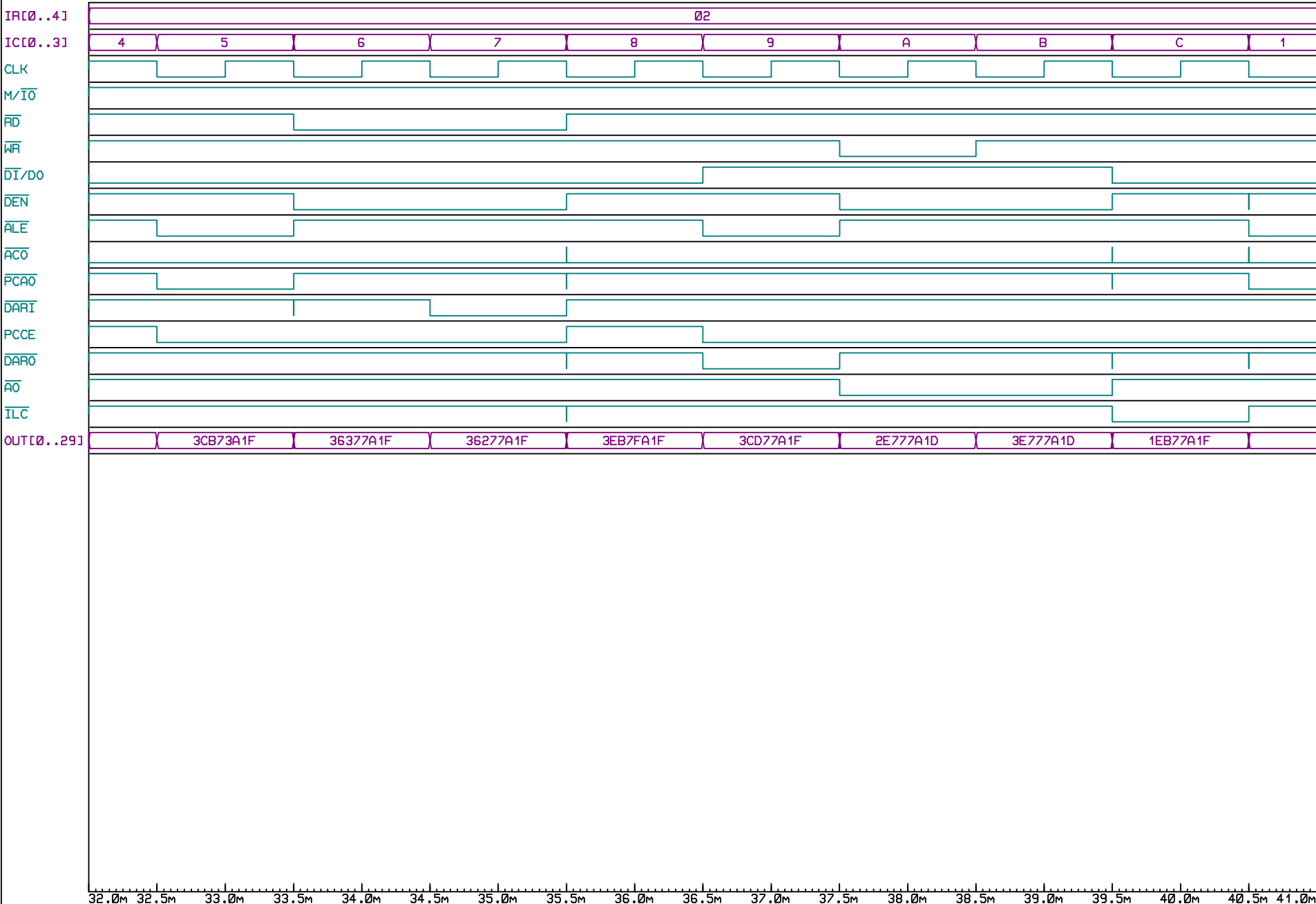
Control Unit - Execution LDA



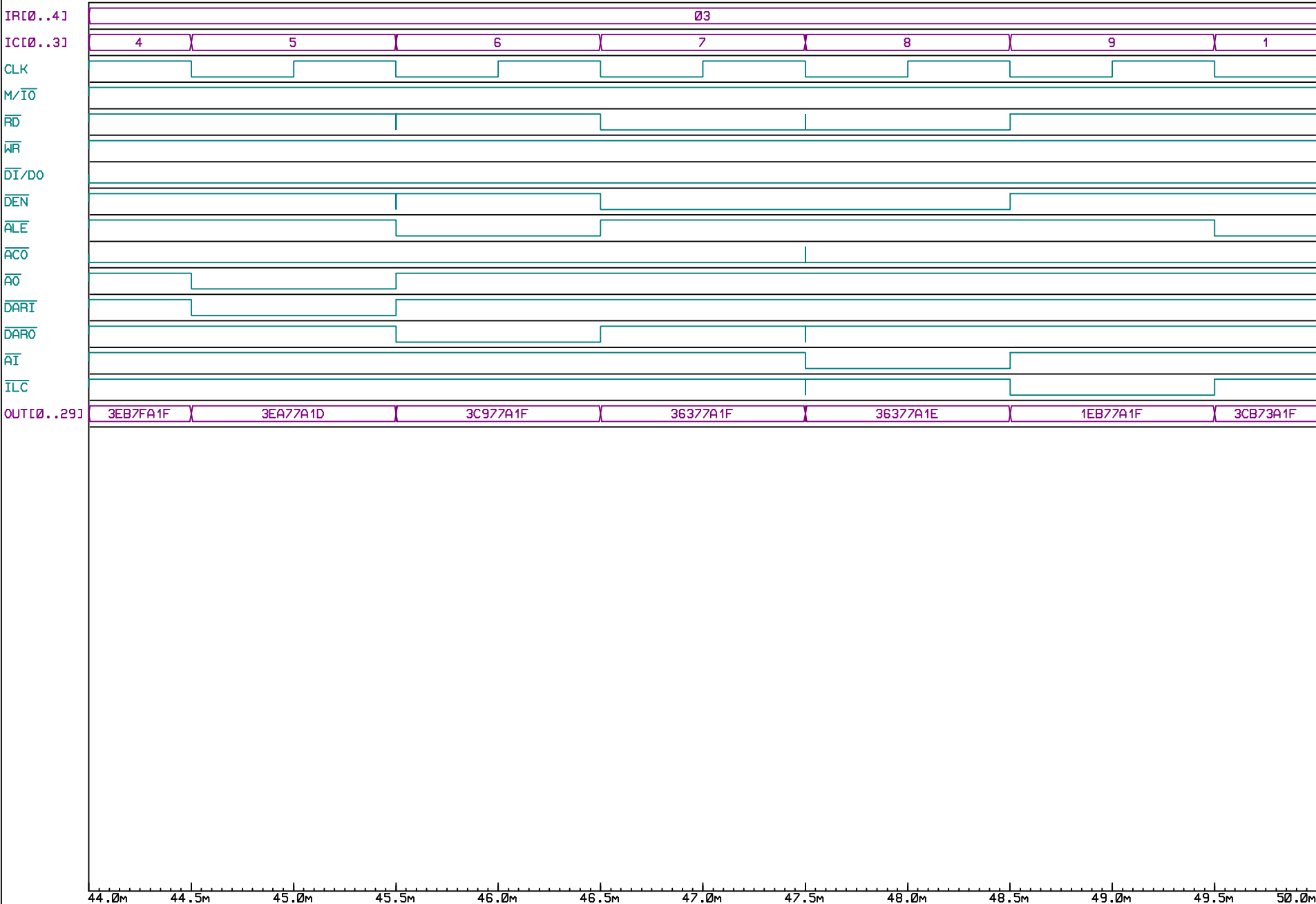
Control Unit - Execution LDI



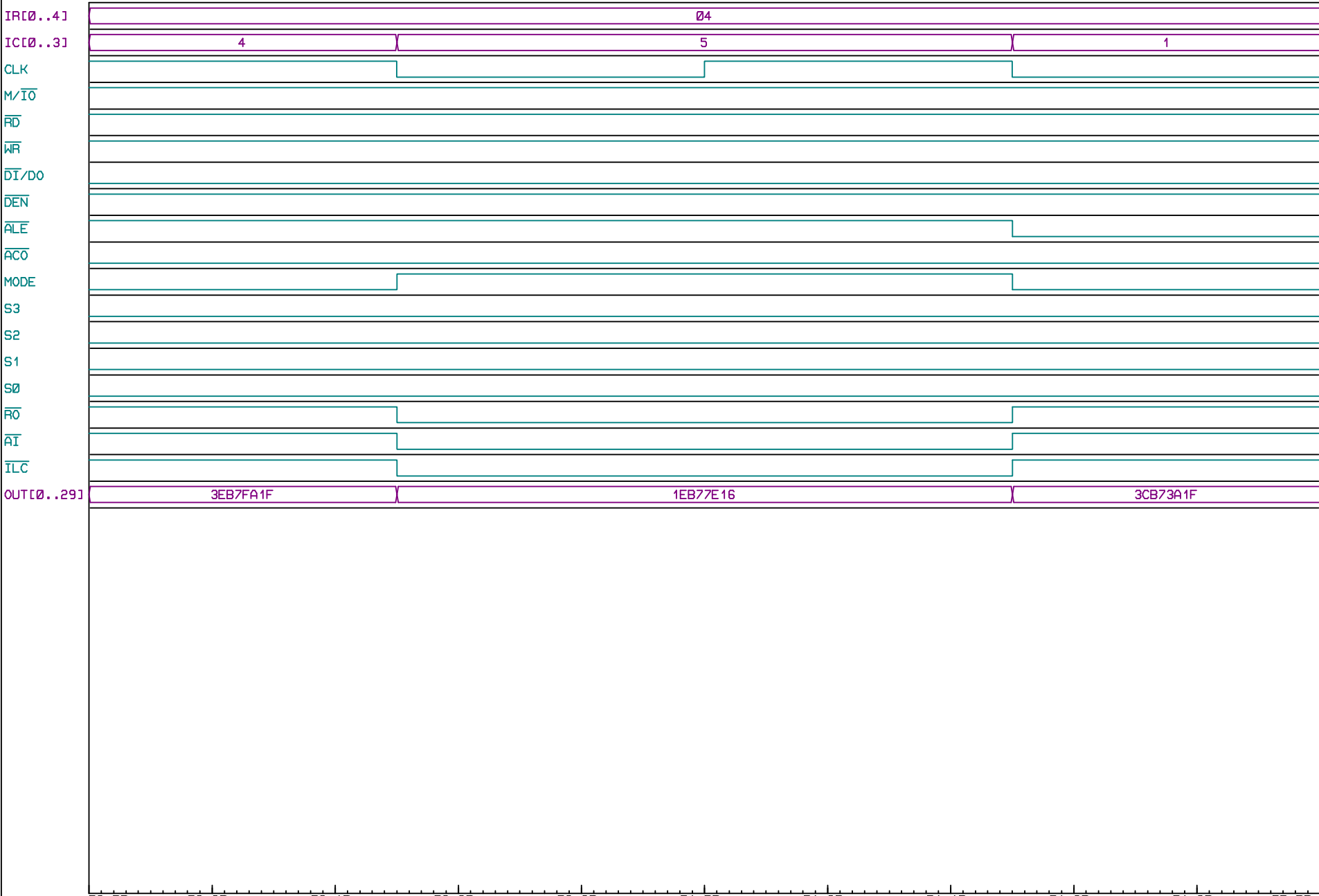
Control Unit - Execution STA



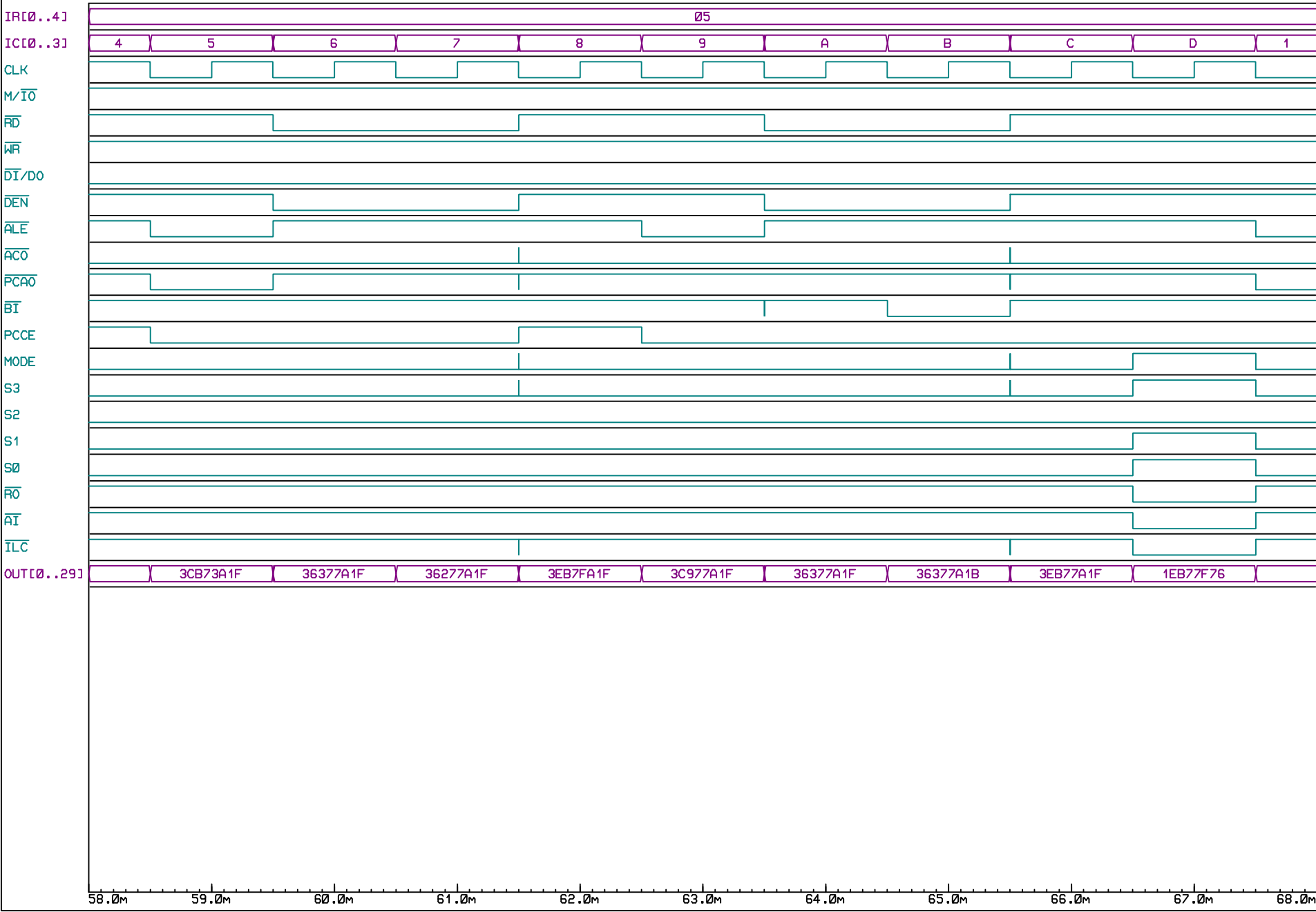
Control Unit - Execution LIA



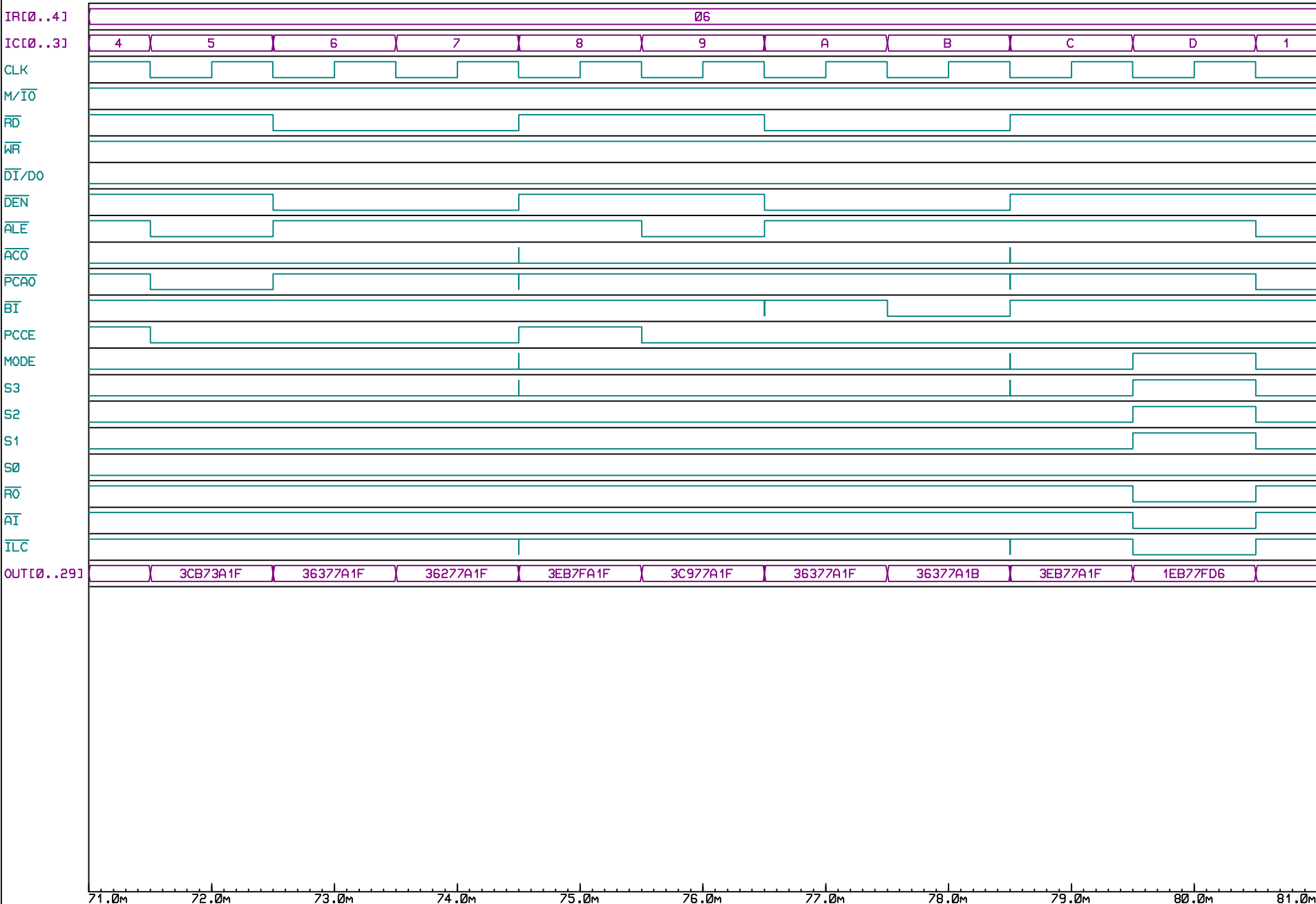
Control Unit - Execution NOT



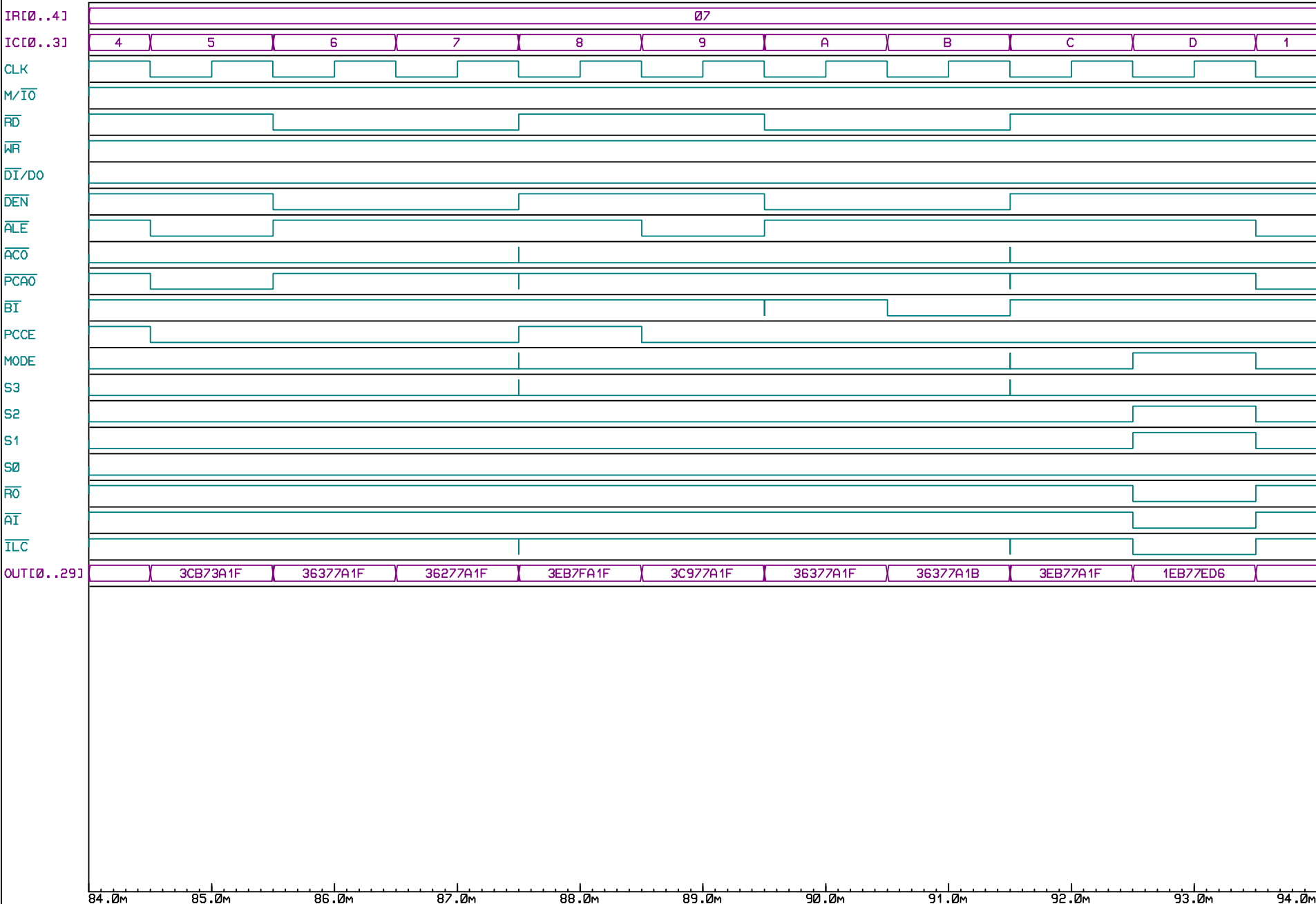
Control Unit - Execution AND



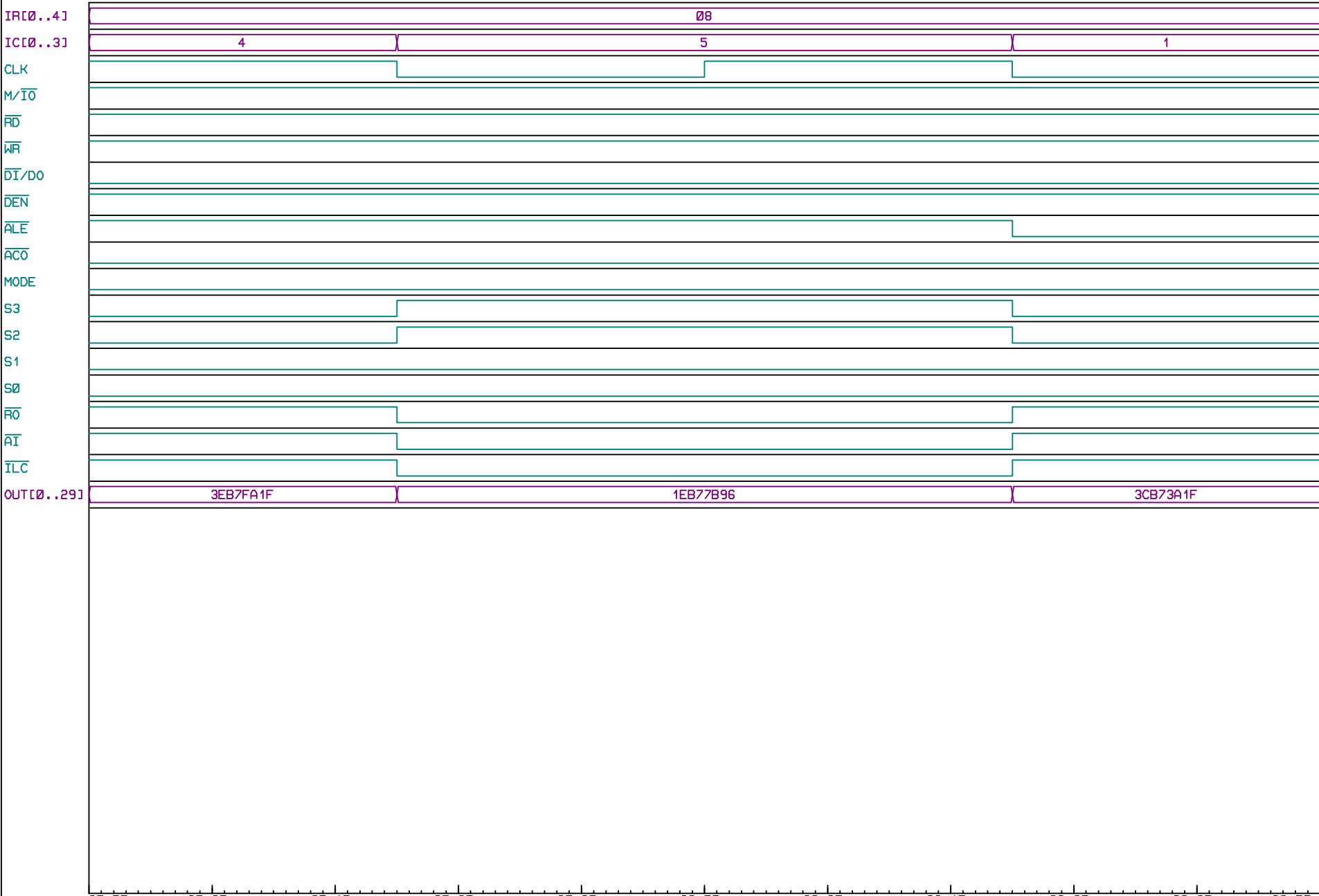
Control Unit - Execution OR



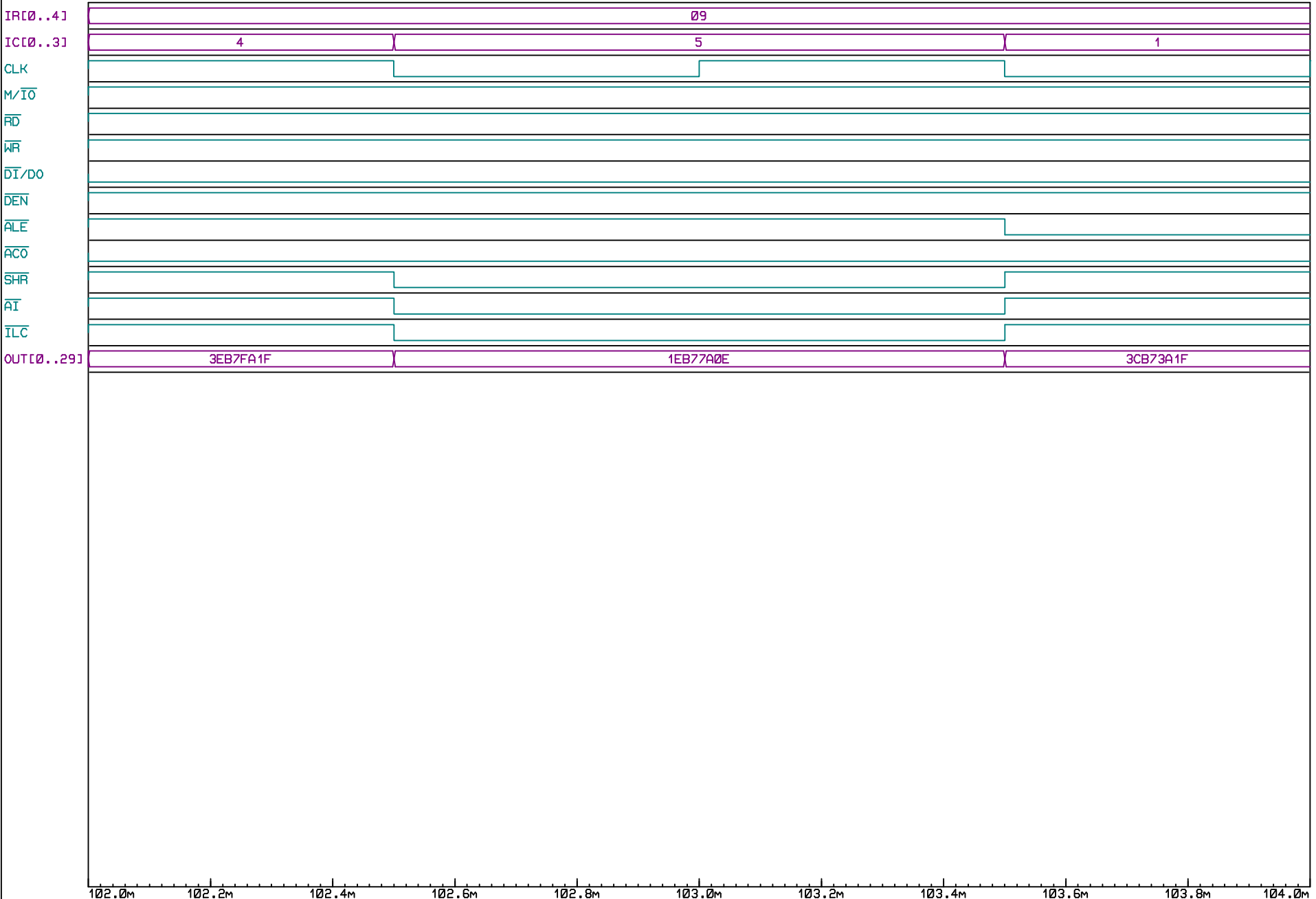
Control Unit - Execution XOR



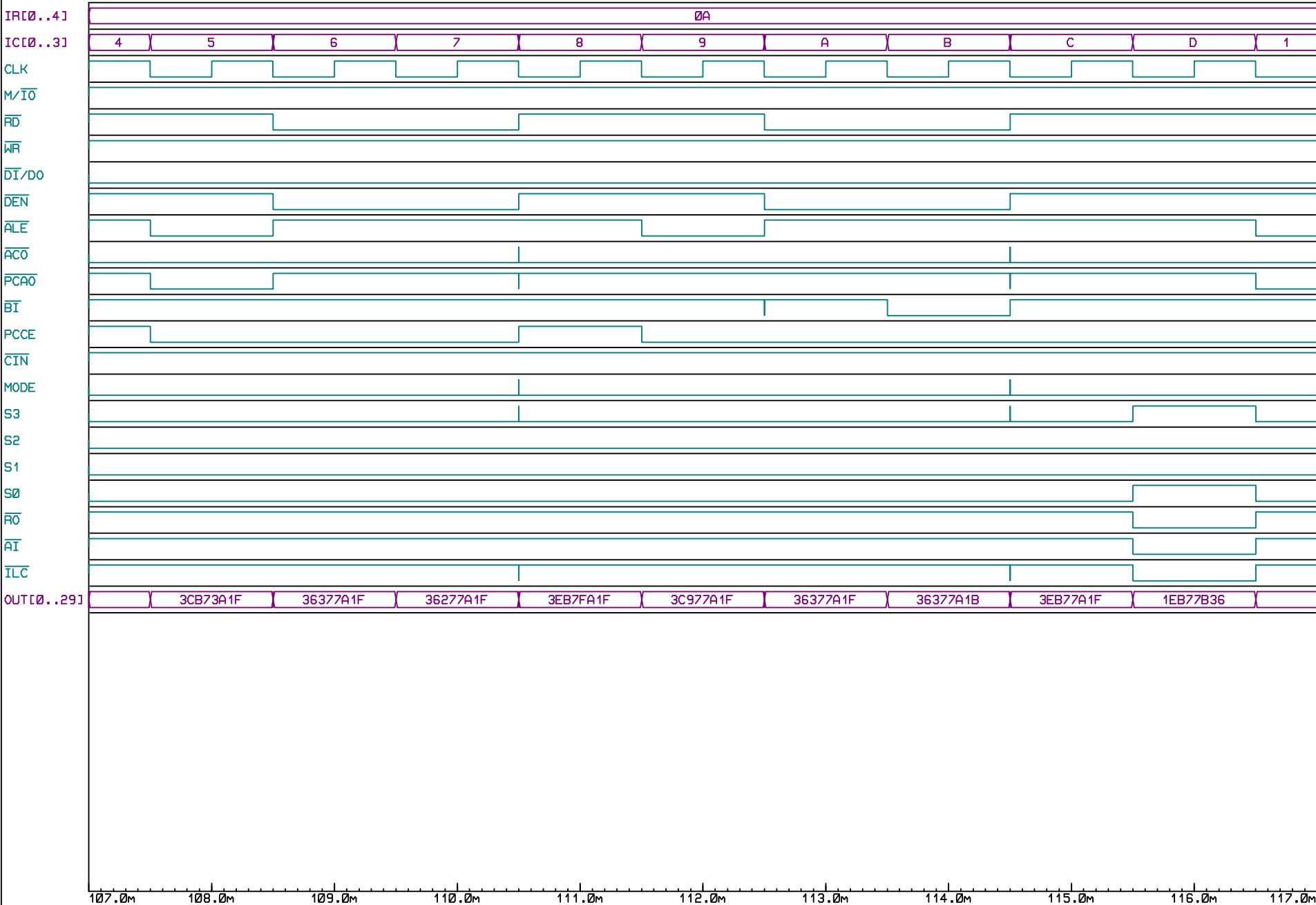
Control Unit - Execution SHL



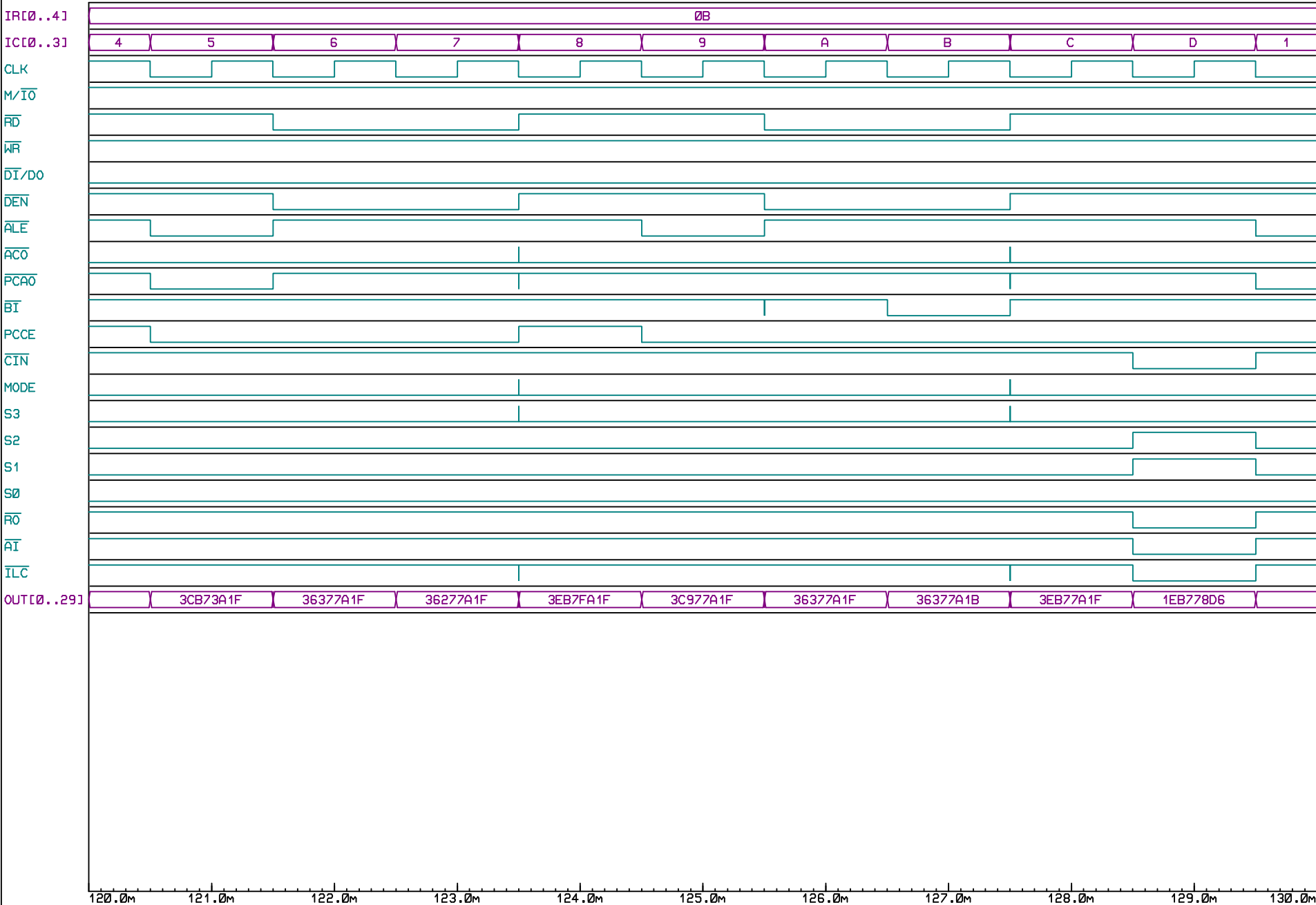
Control Unit - Execution SHR



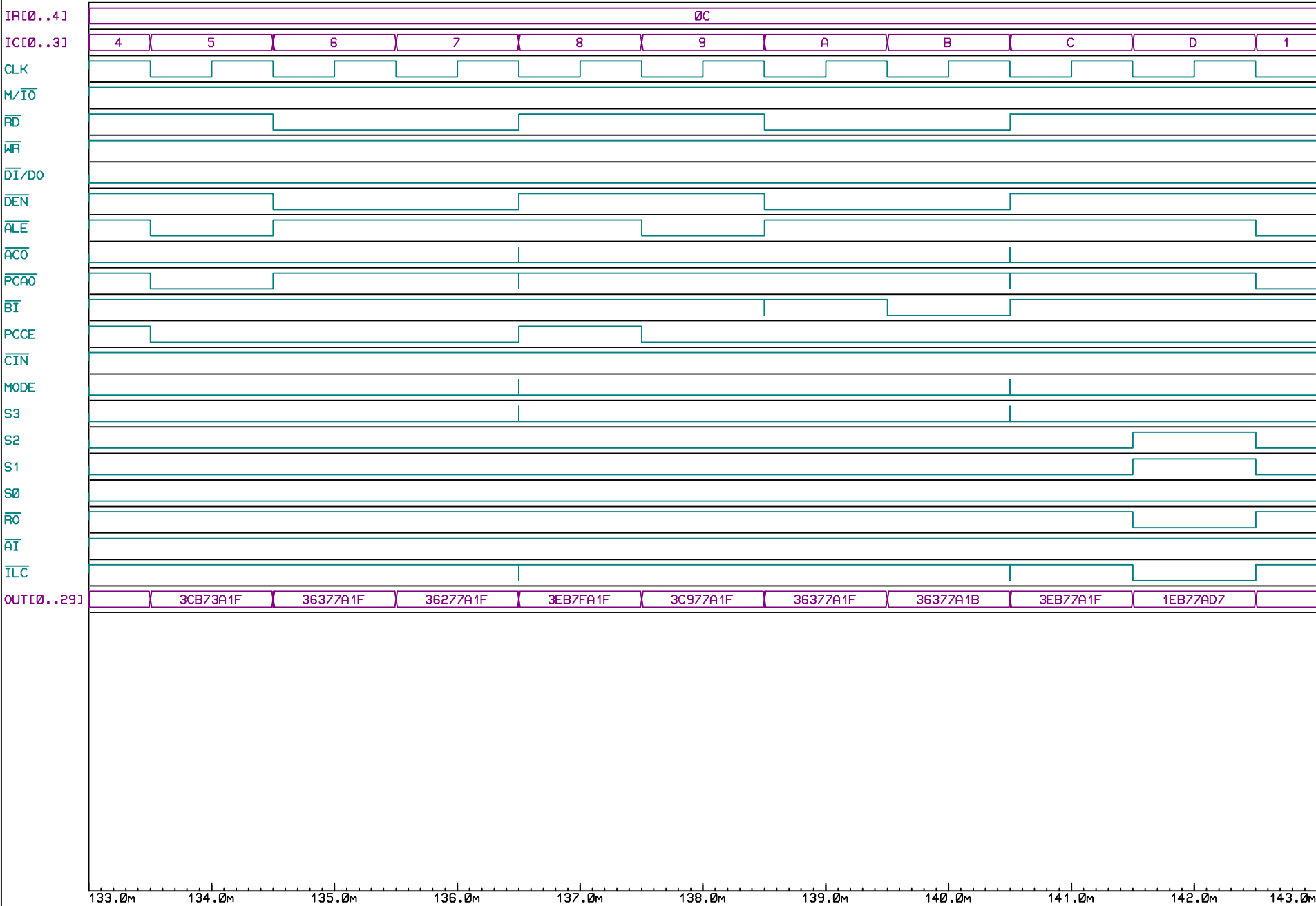
Control Unit - Execution ADD



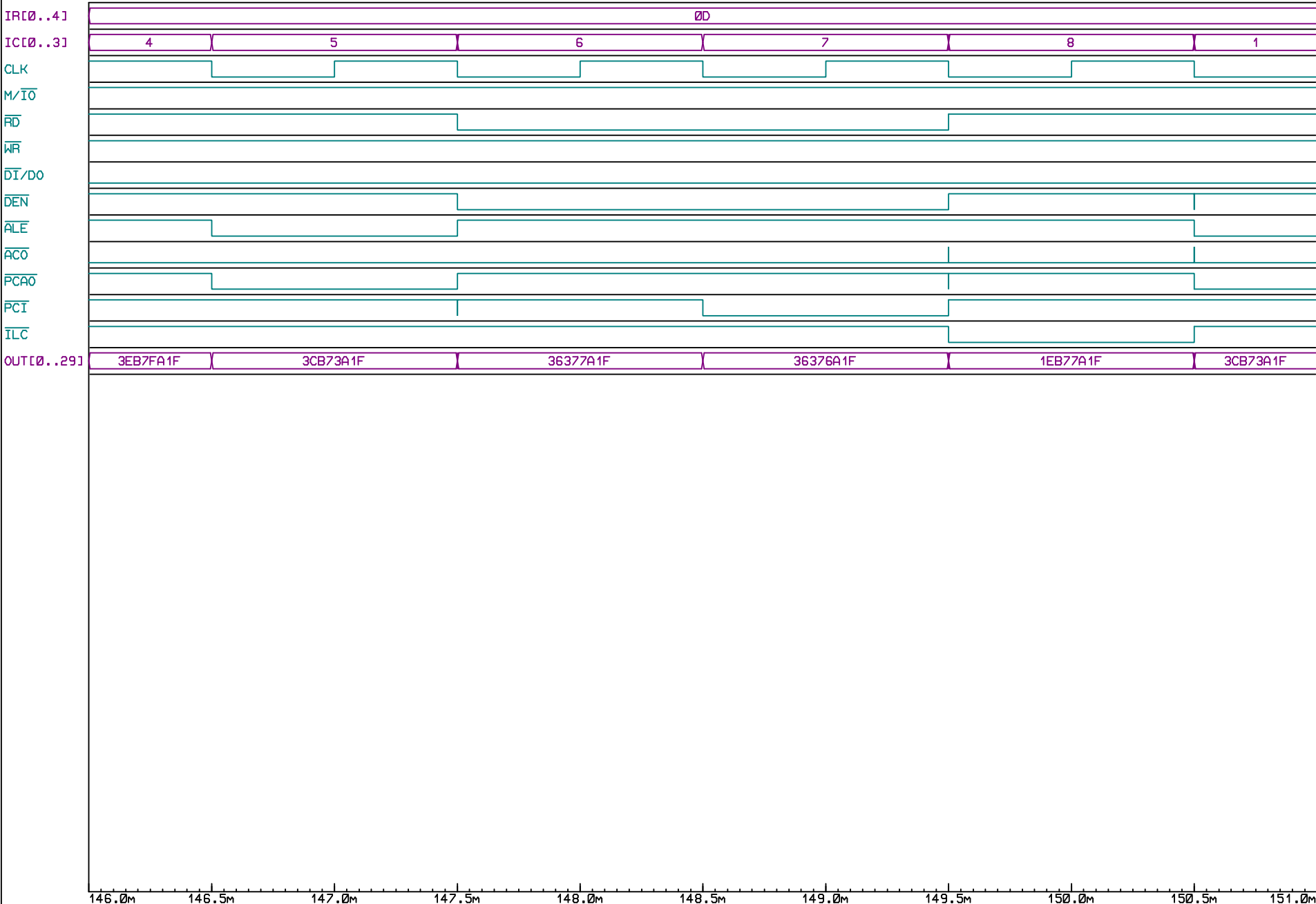
Control Unit - Execution SUB



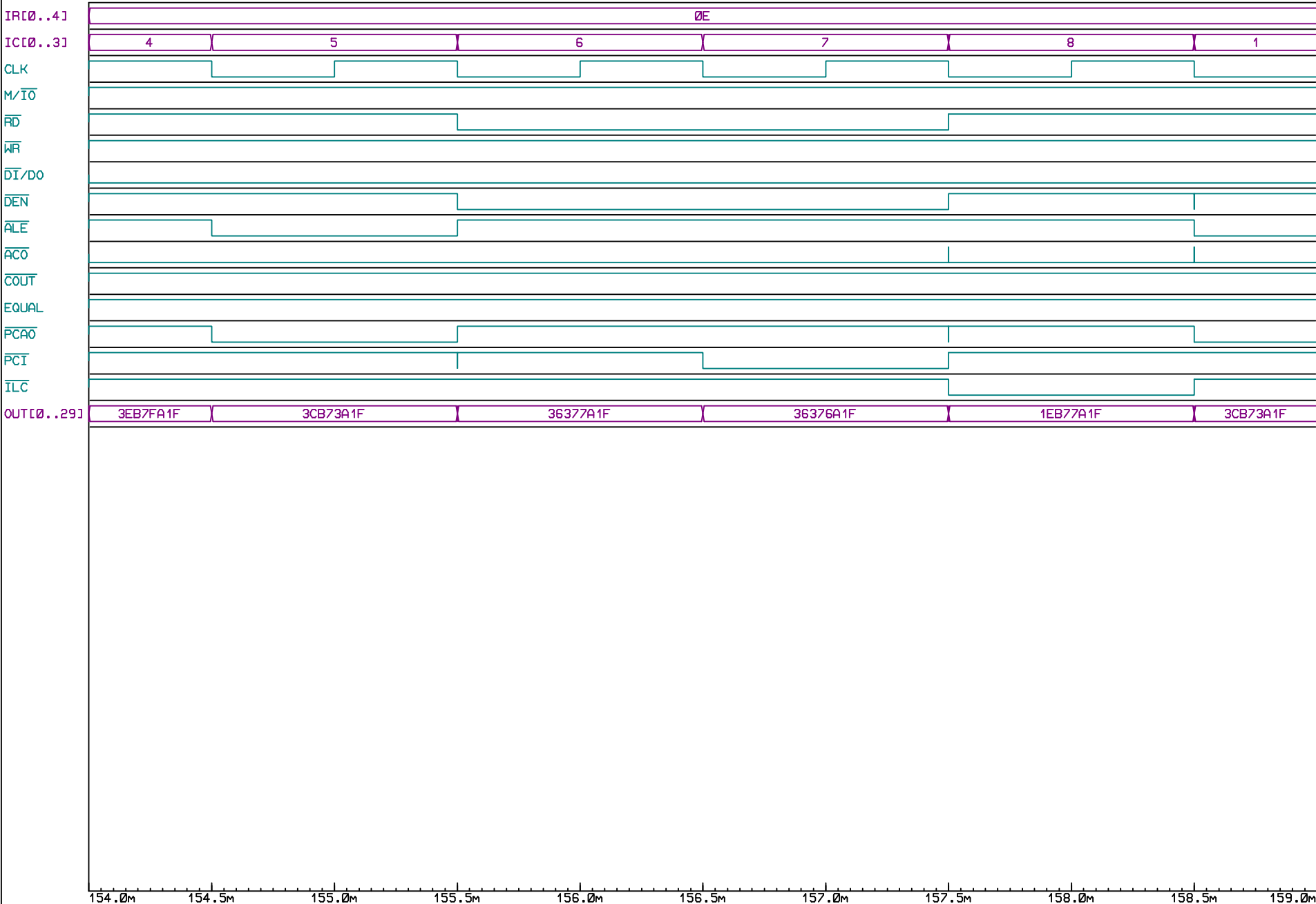
Control Unit - Execution CMP



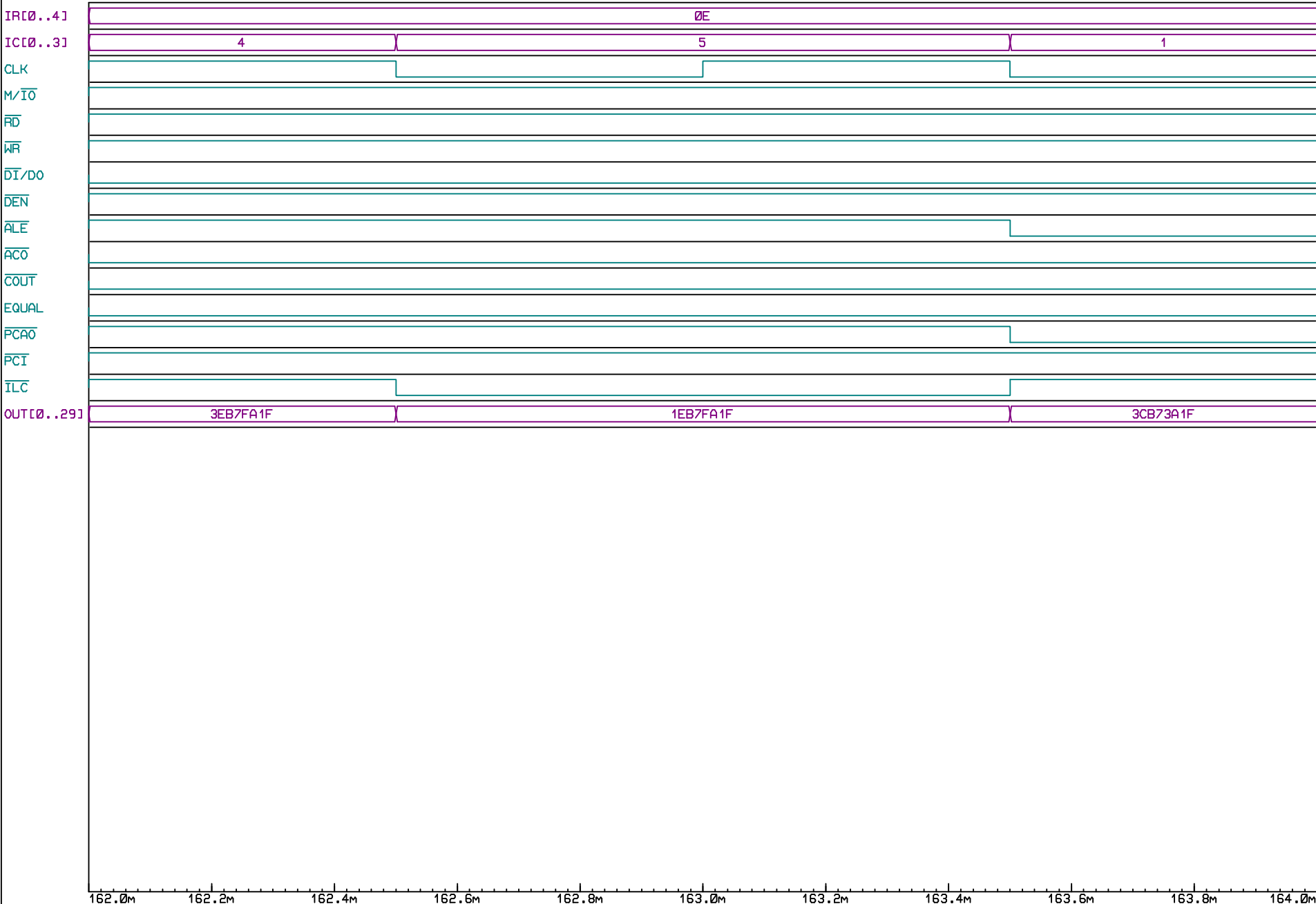
Control Unit - Execution JMP



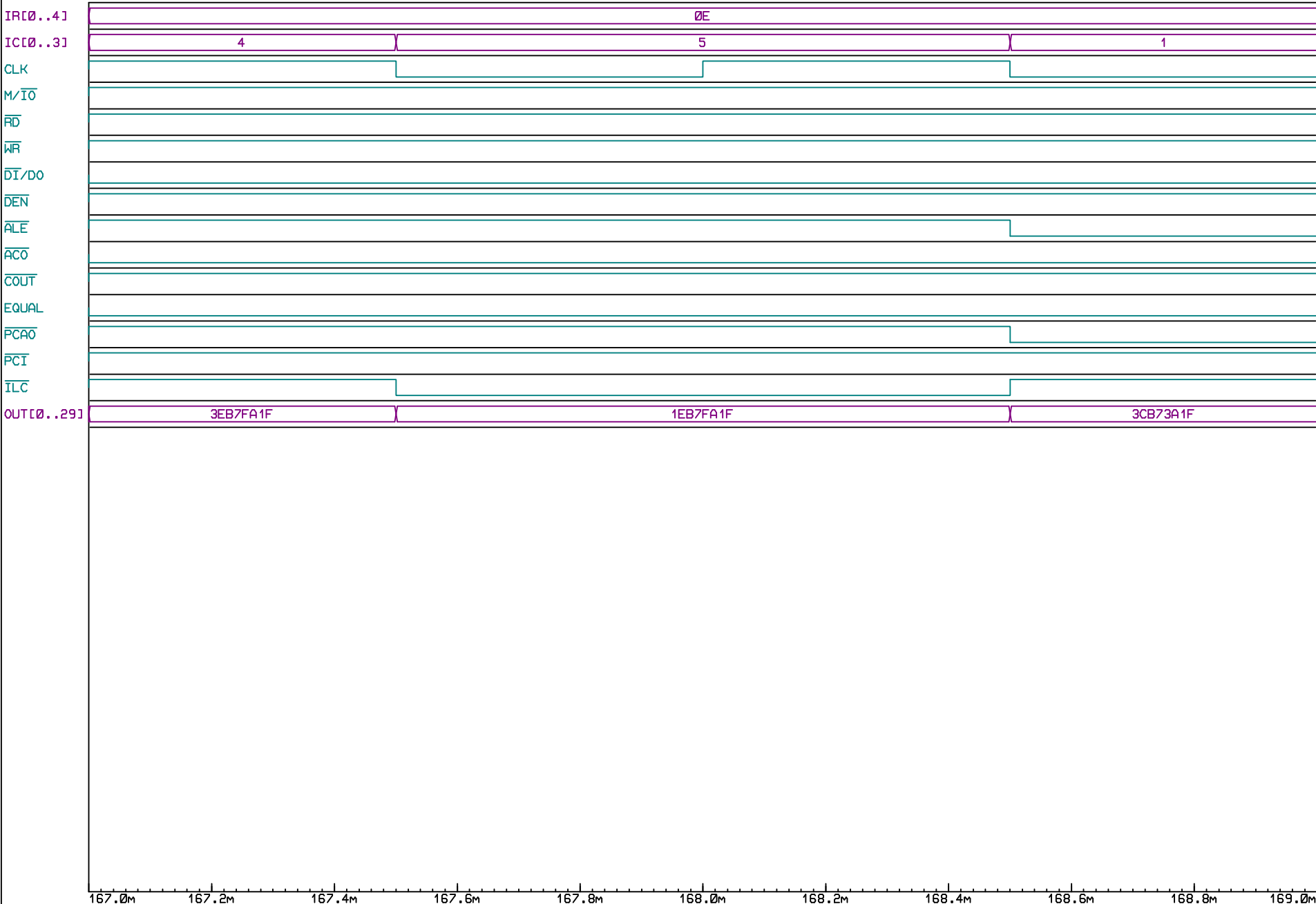
Control Unit - Exec JPE (A=B)



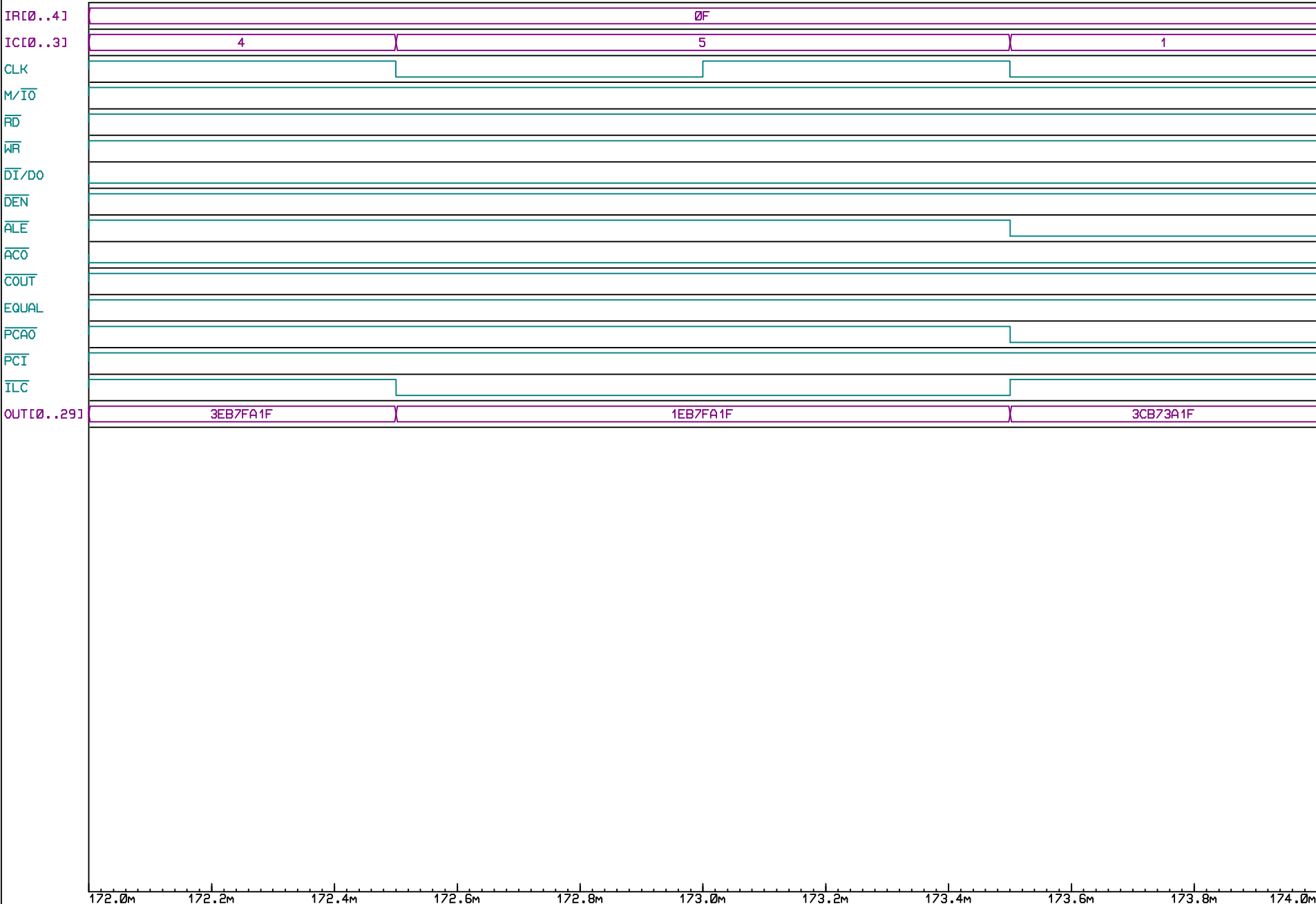
Control Unit - Exec JPE (A>B)



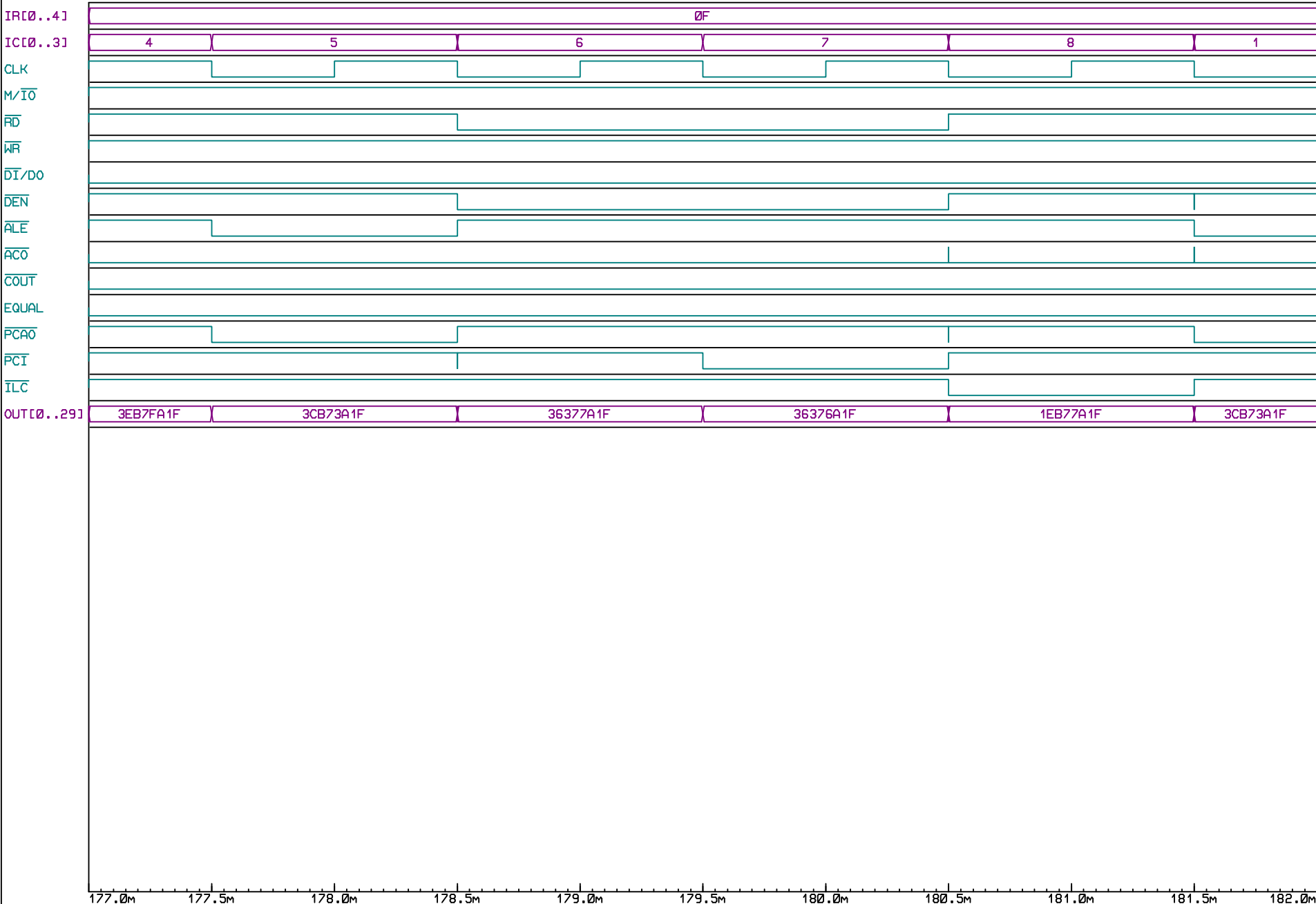
Control Unit - Exec JPE (A<B)



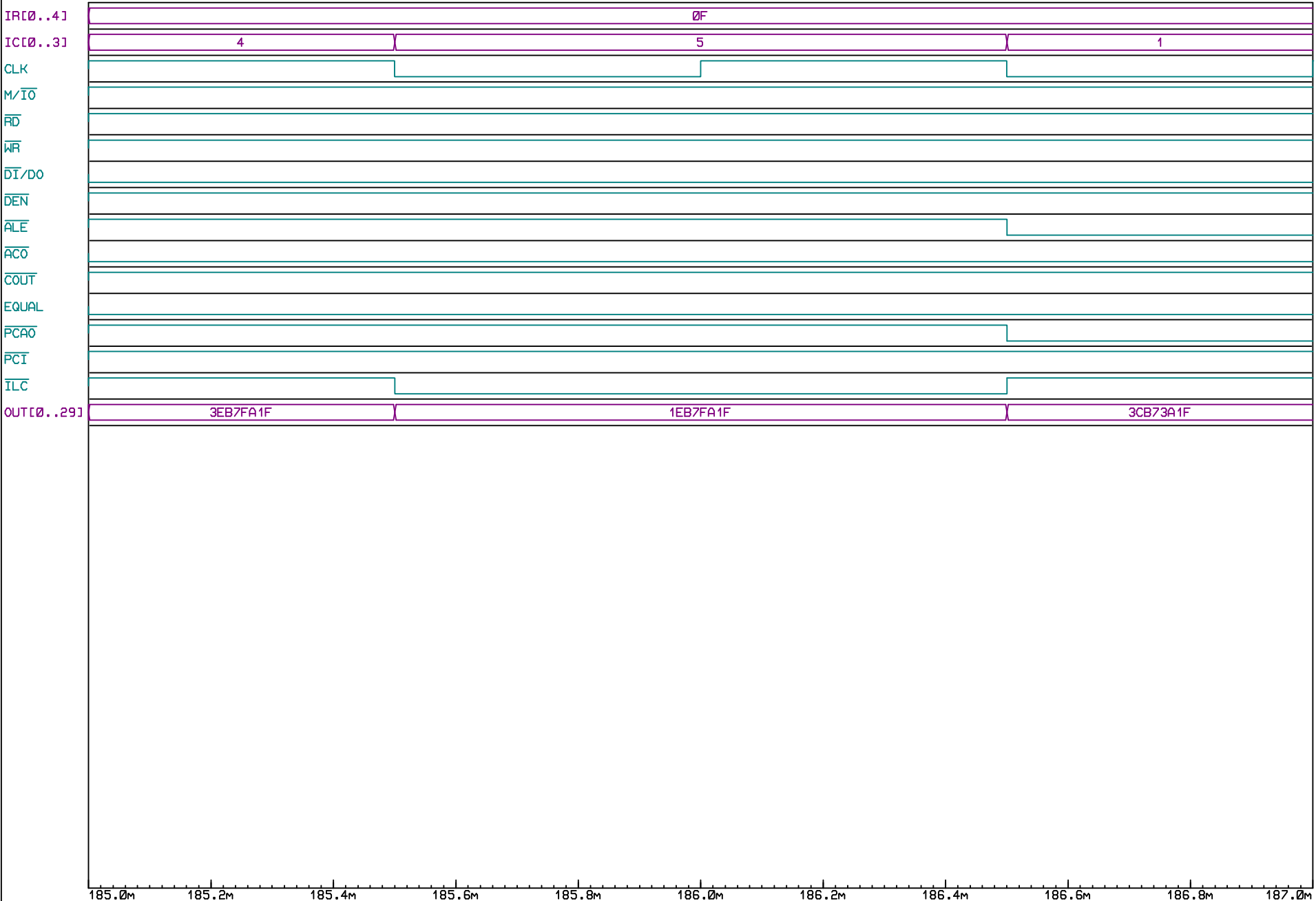
Control Unit - Exec JPA (A=B)



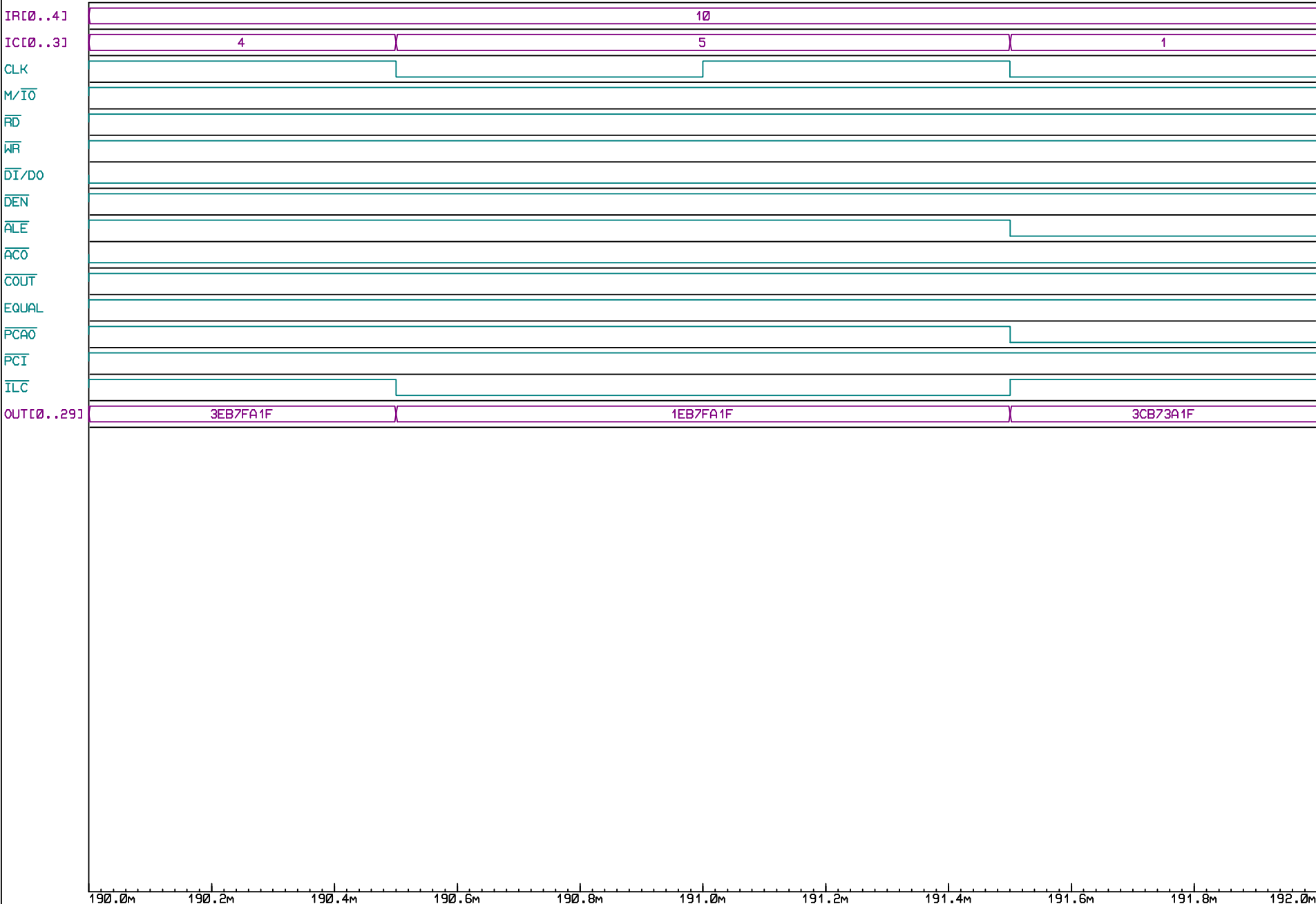
Control Unit - Exec JPA (A>B)



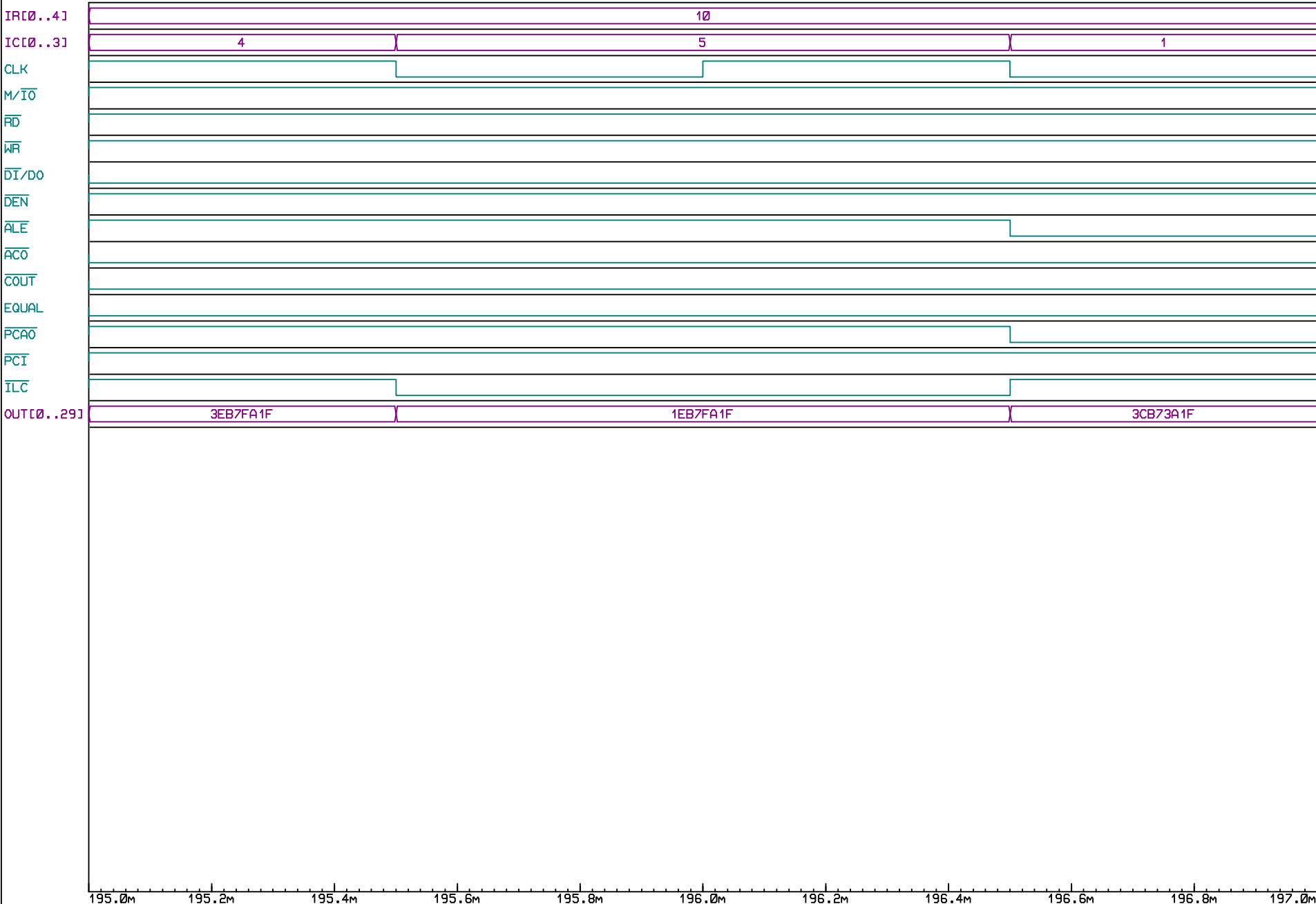
Control Unit - Exec JPA (A<B)



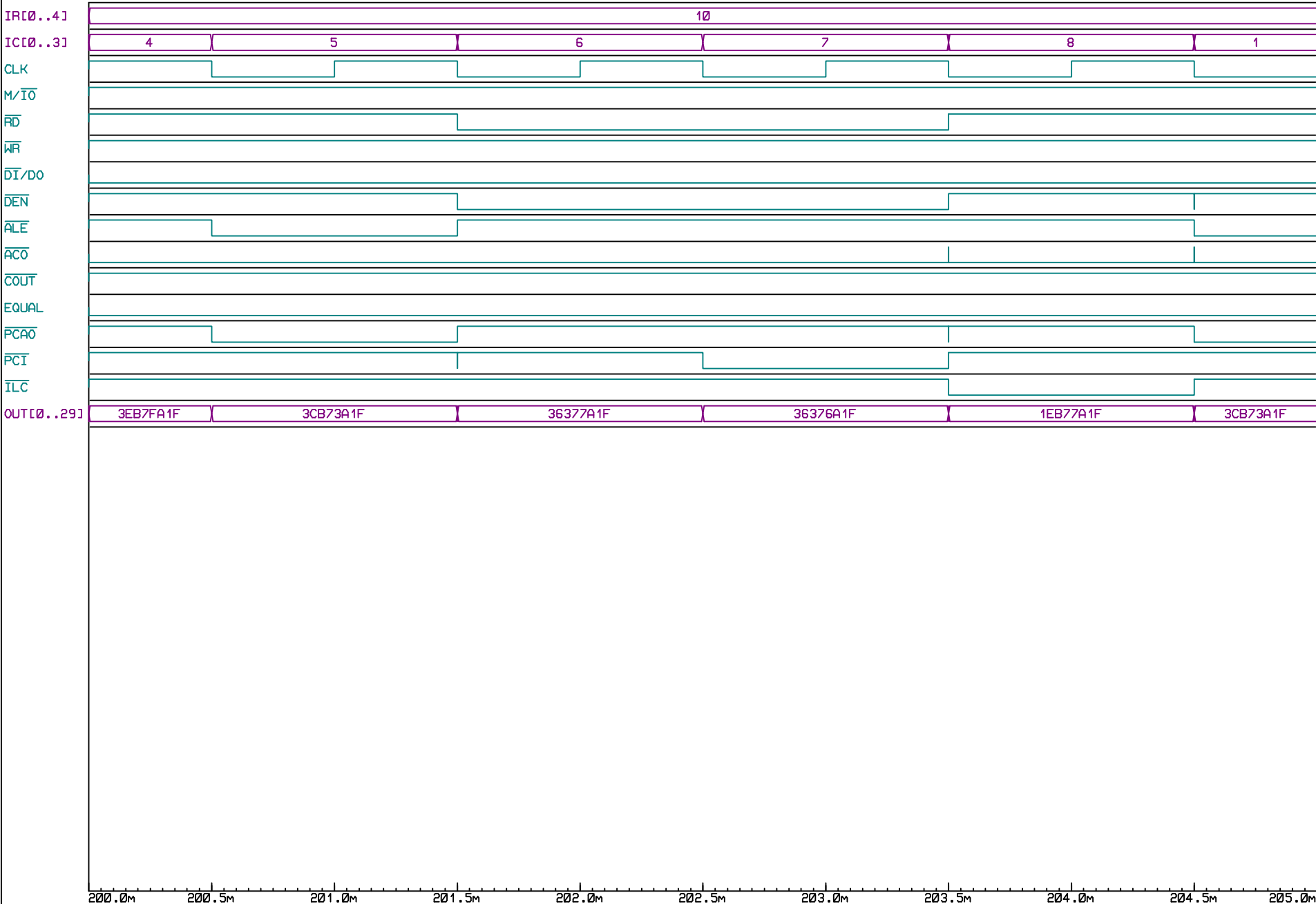
Control Unit - Exec JPB (A=B)



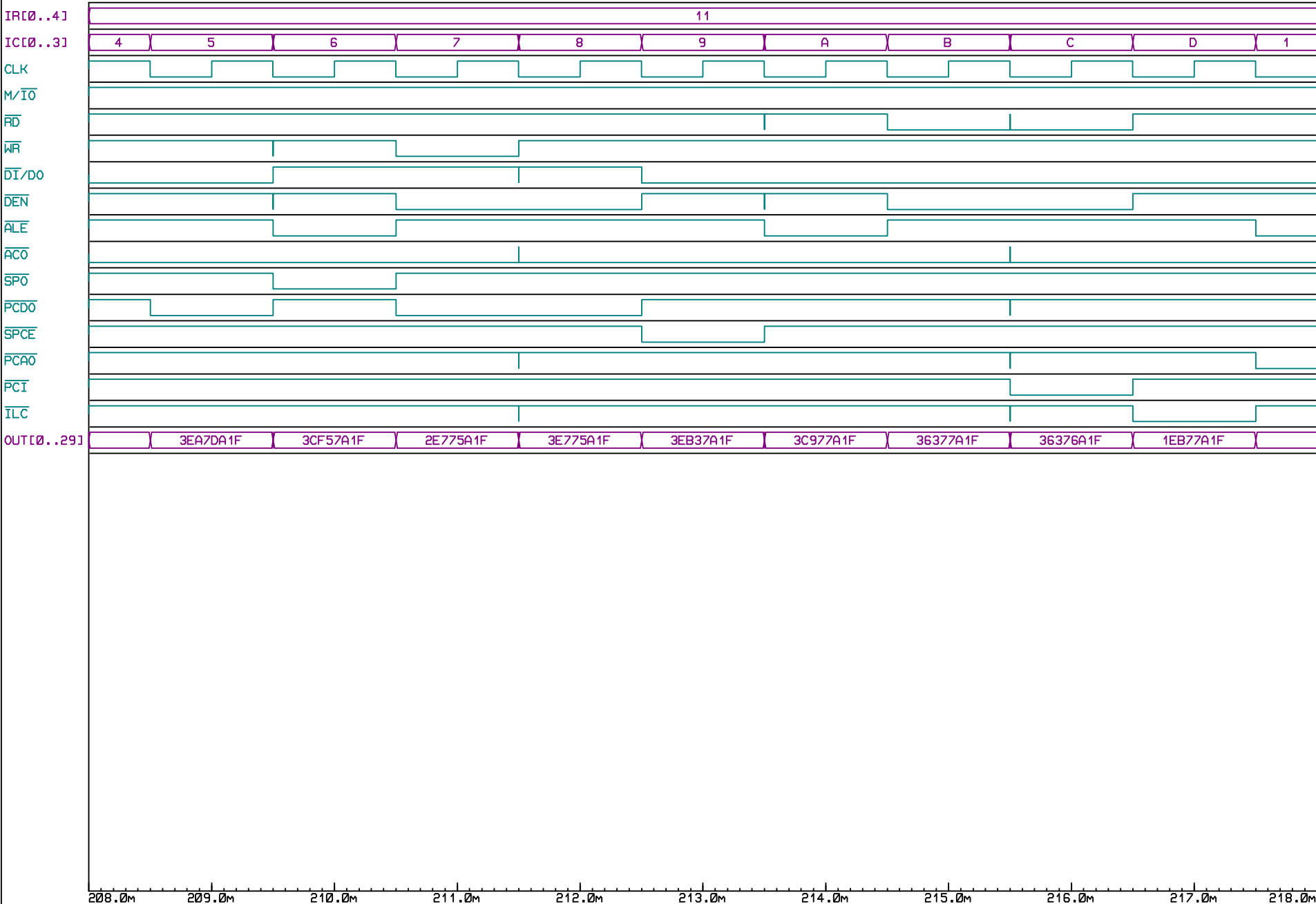
Control Unit - Exec JPB (A>B)



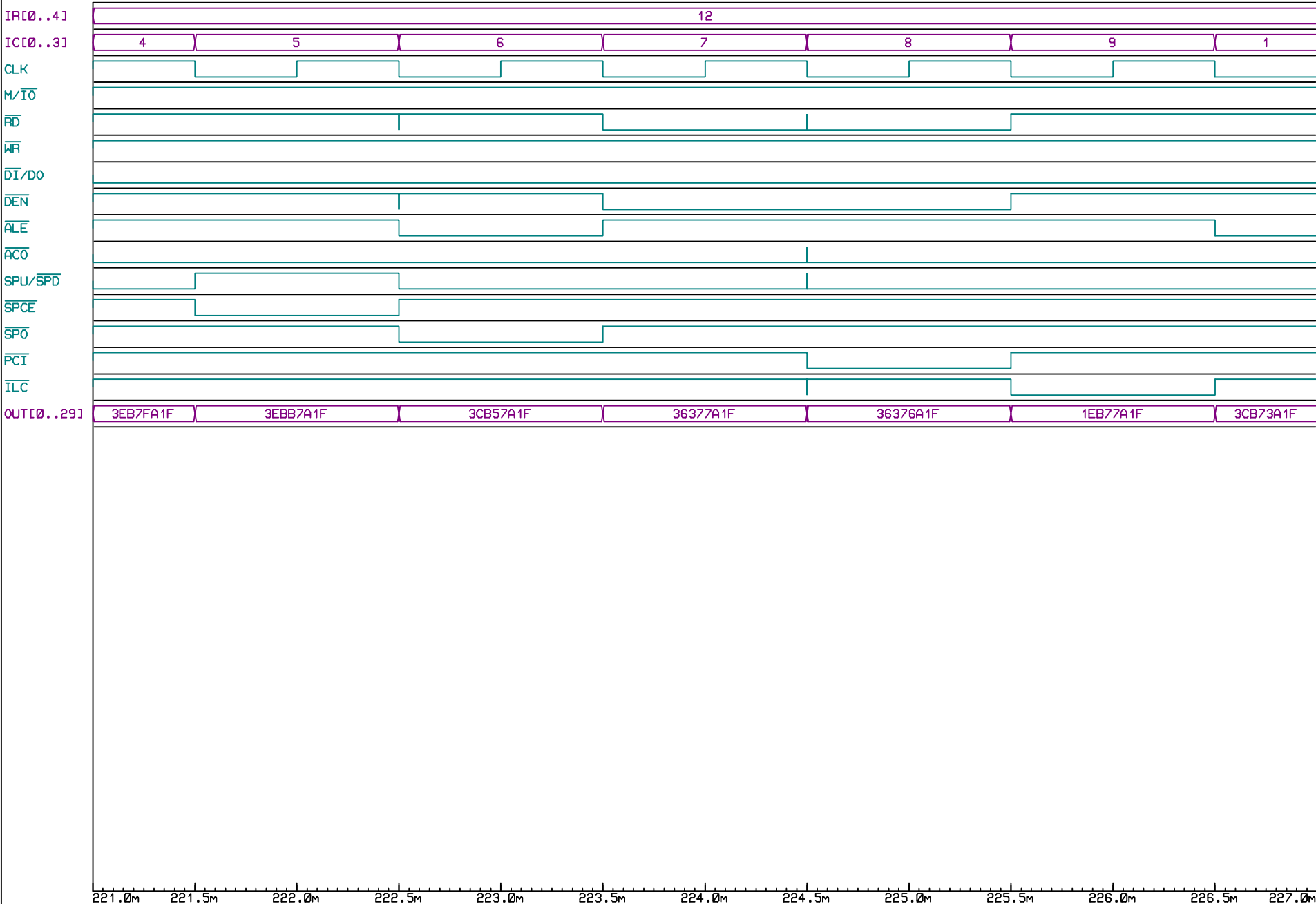
Control Unit - Exec JPB (A<B)



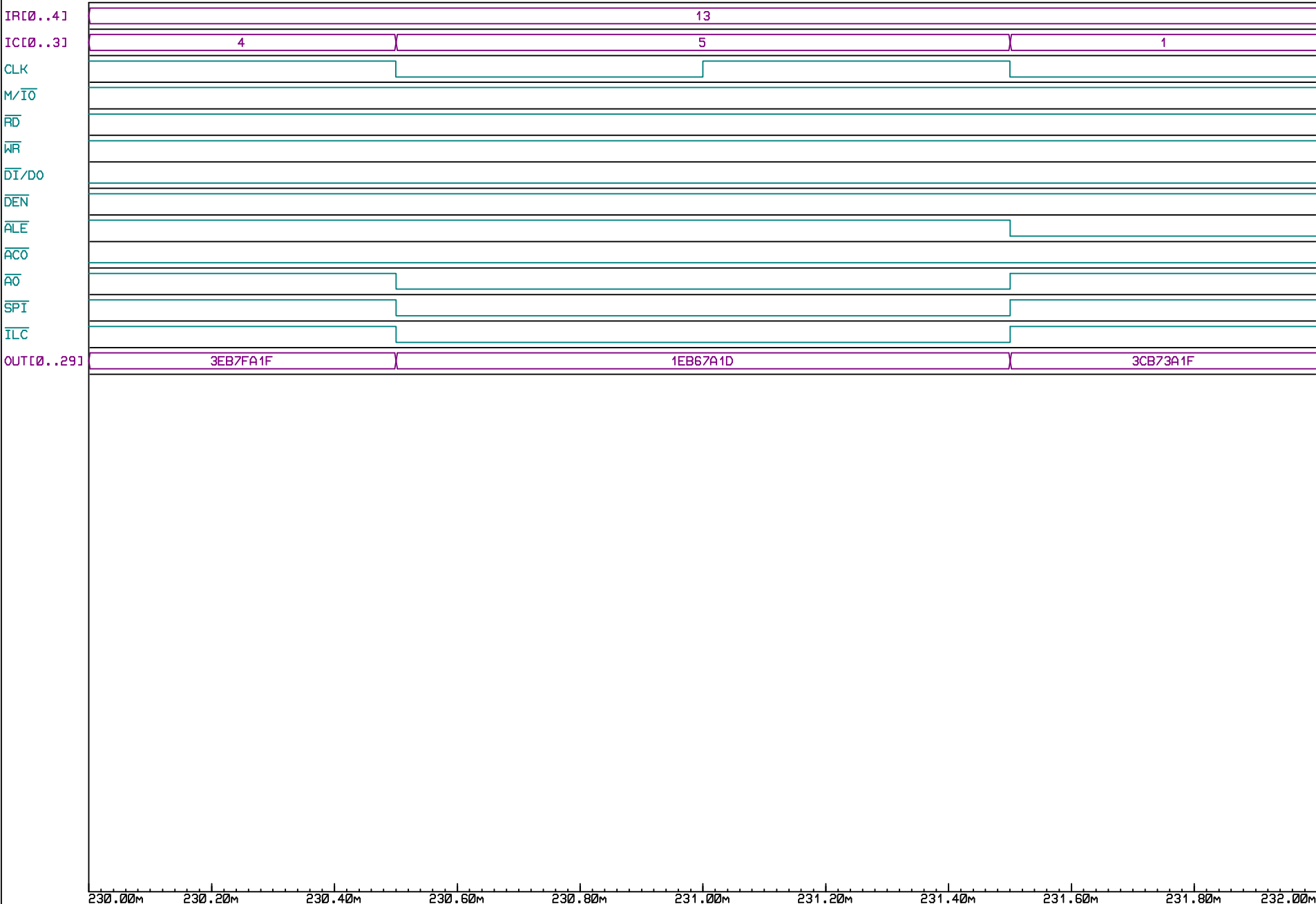
Control Unit - Execution CALL



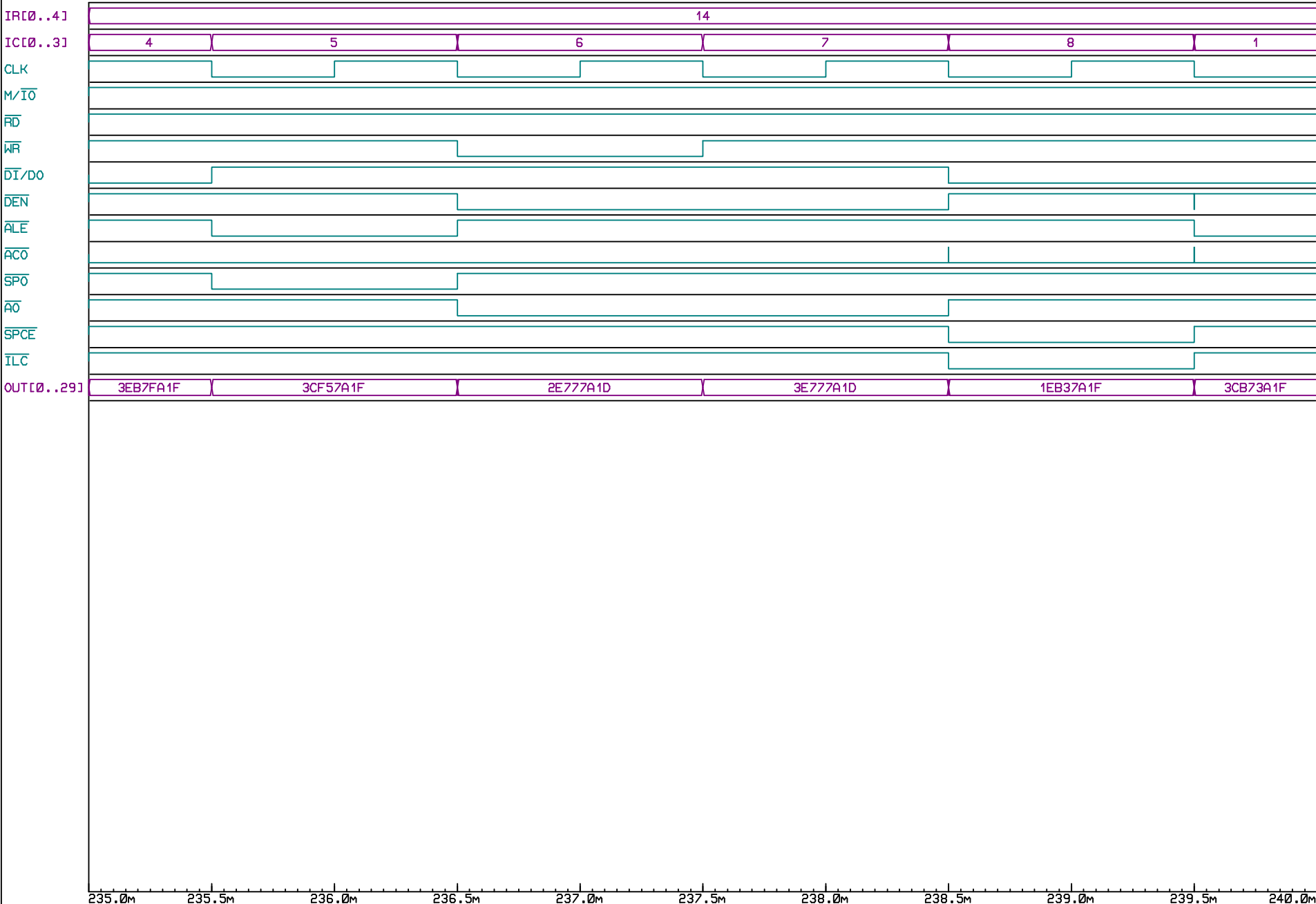
Control Unit - Execution RET



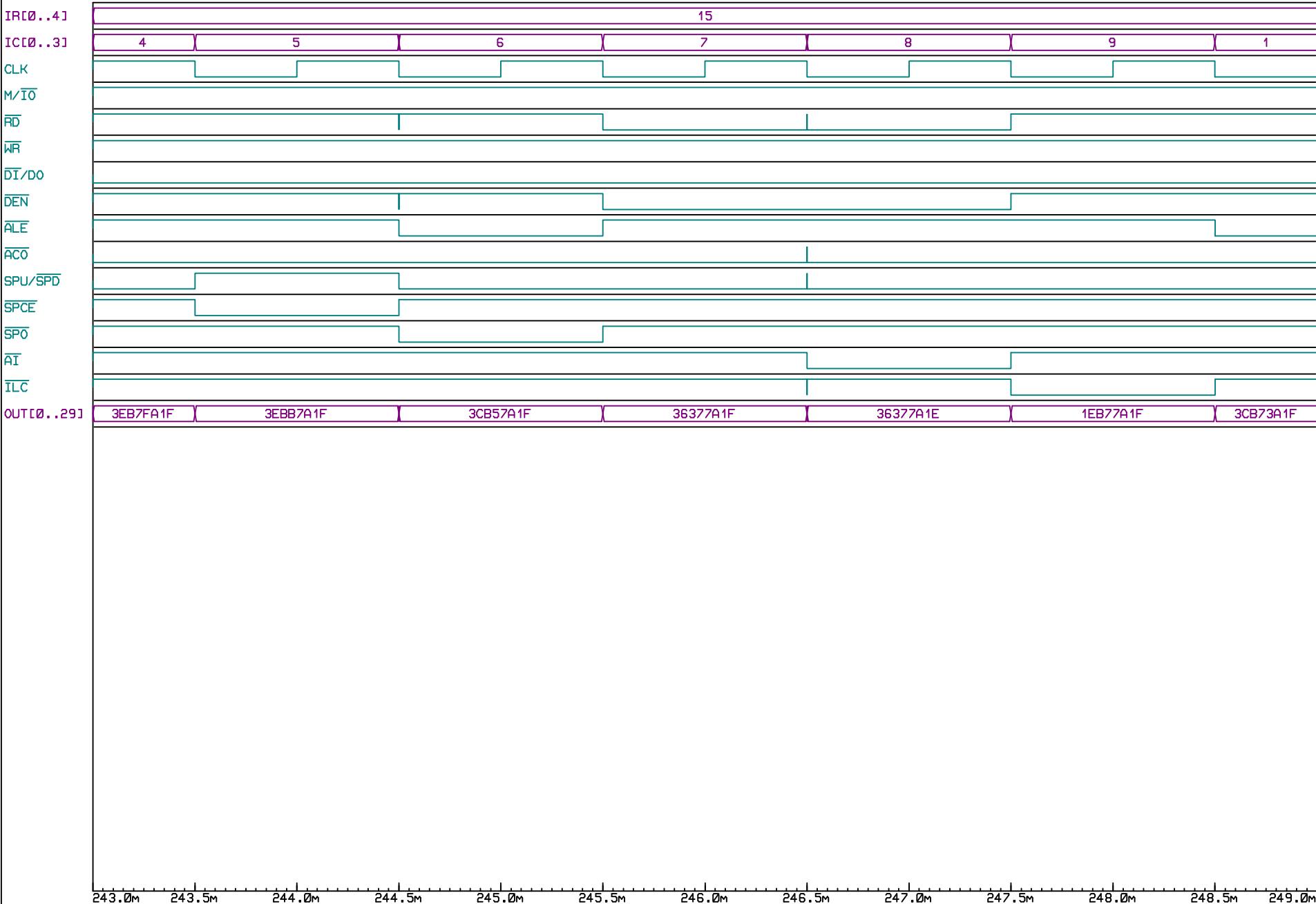
Control Unit - Execution LSP



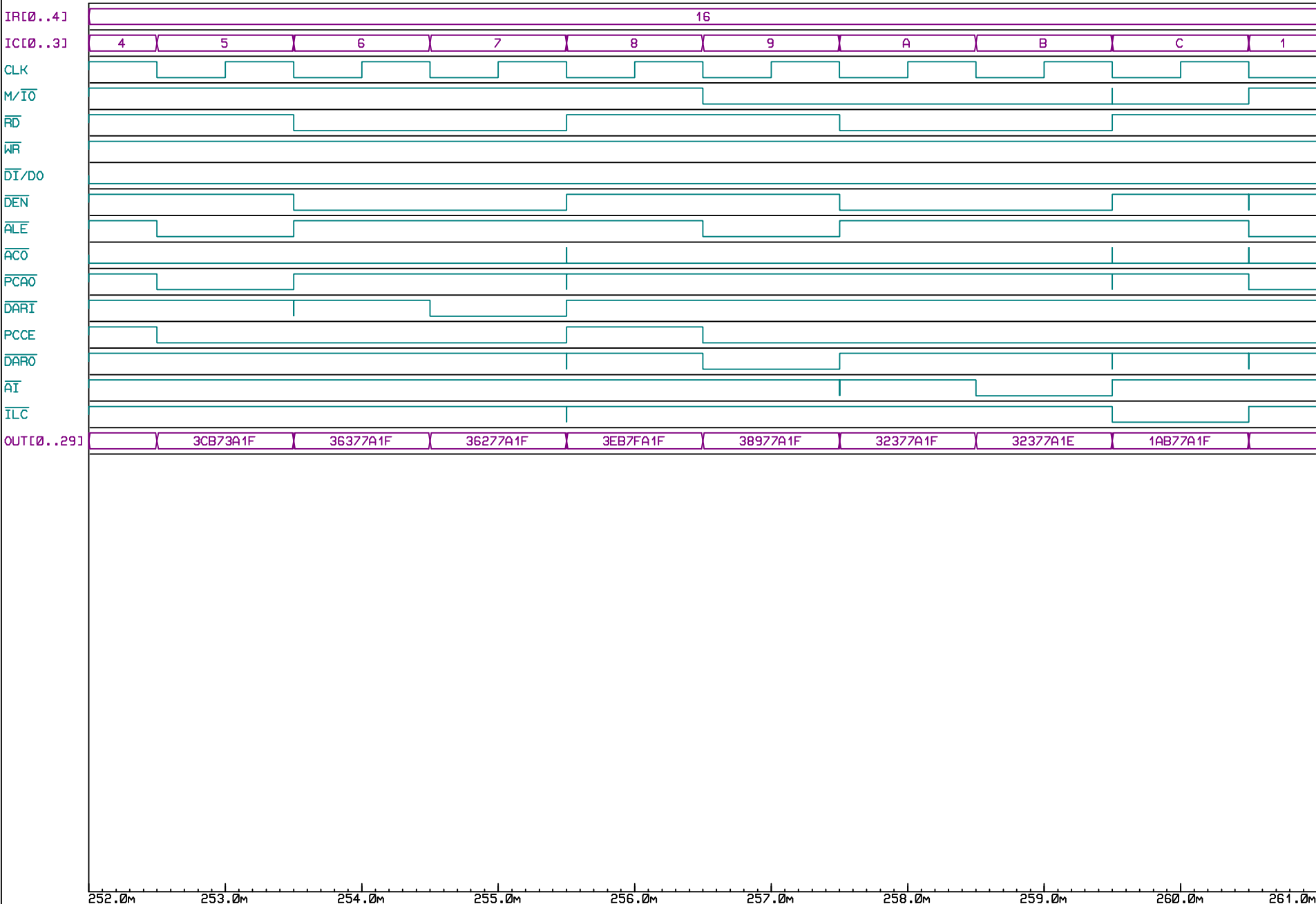
Control Unit - Execution PUSH



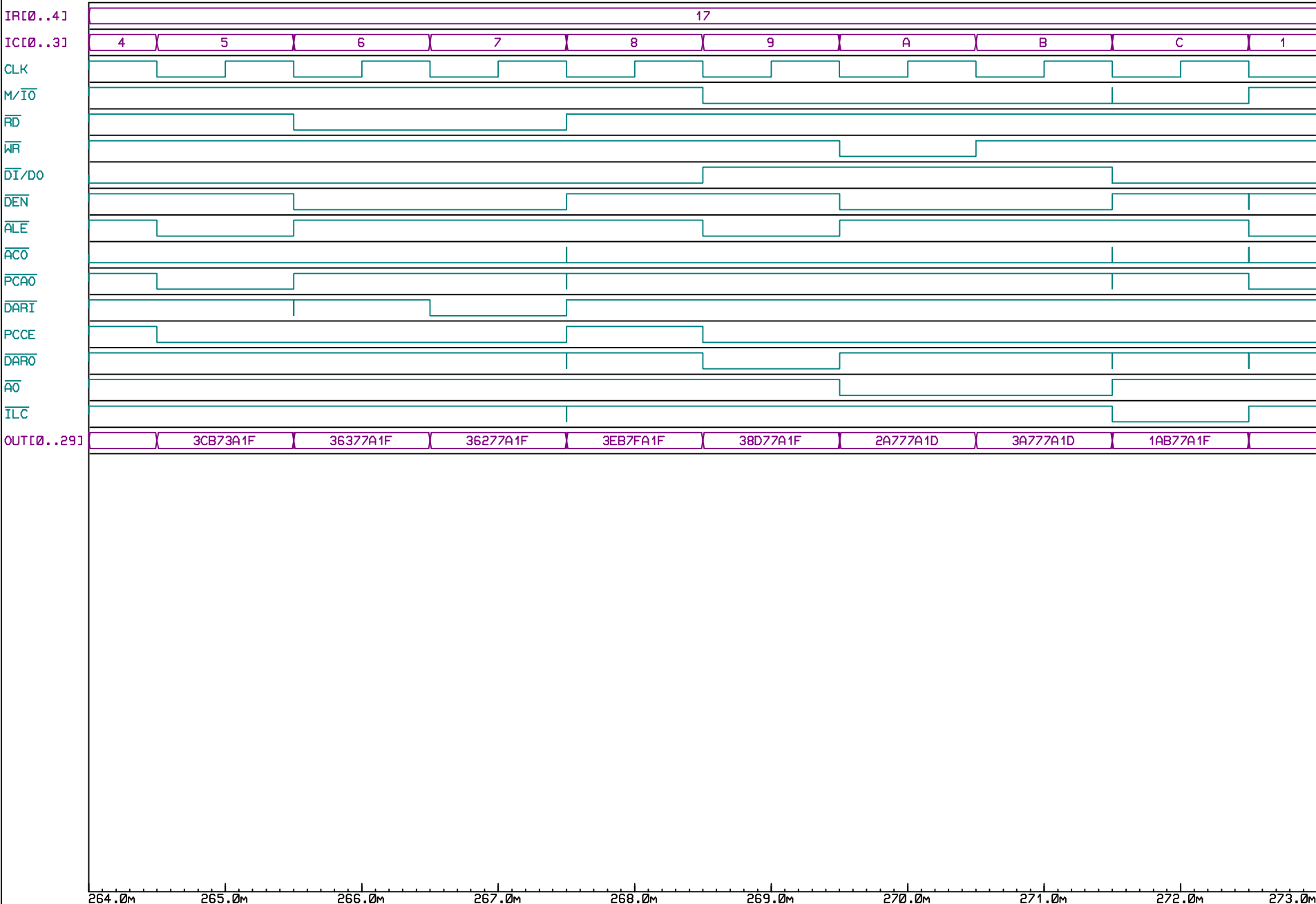
Control Unit - Execution POP



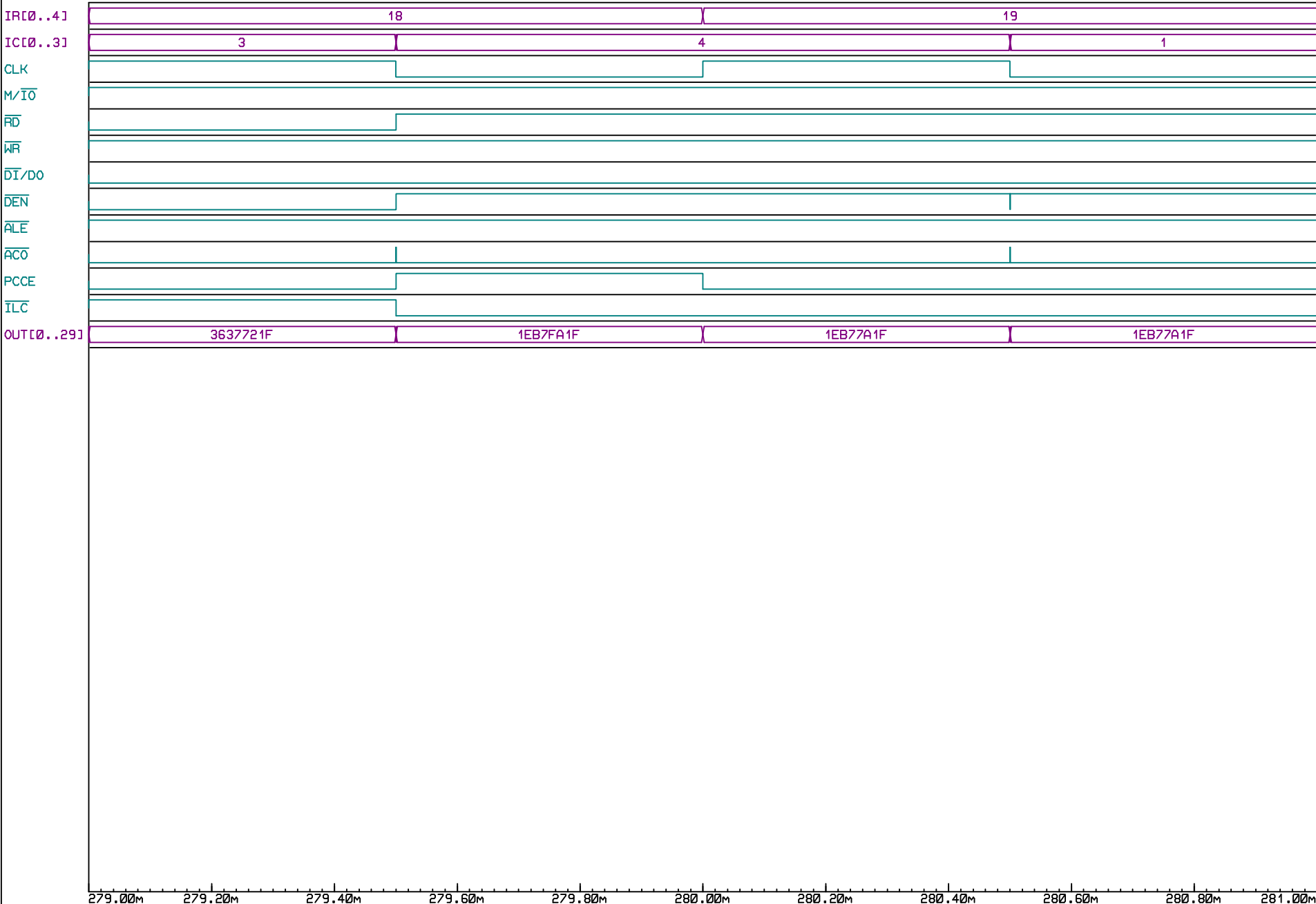
Control Unit - Execution IN



Control Unit - Execution OUT



Control Unit - Execution NOP



Control Unit - Execution HALT

